

MS-EXCEL

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Decade of experience in Corporate Training Classroom sessions in different streams of Technology.

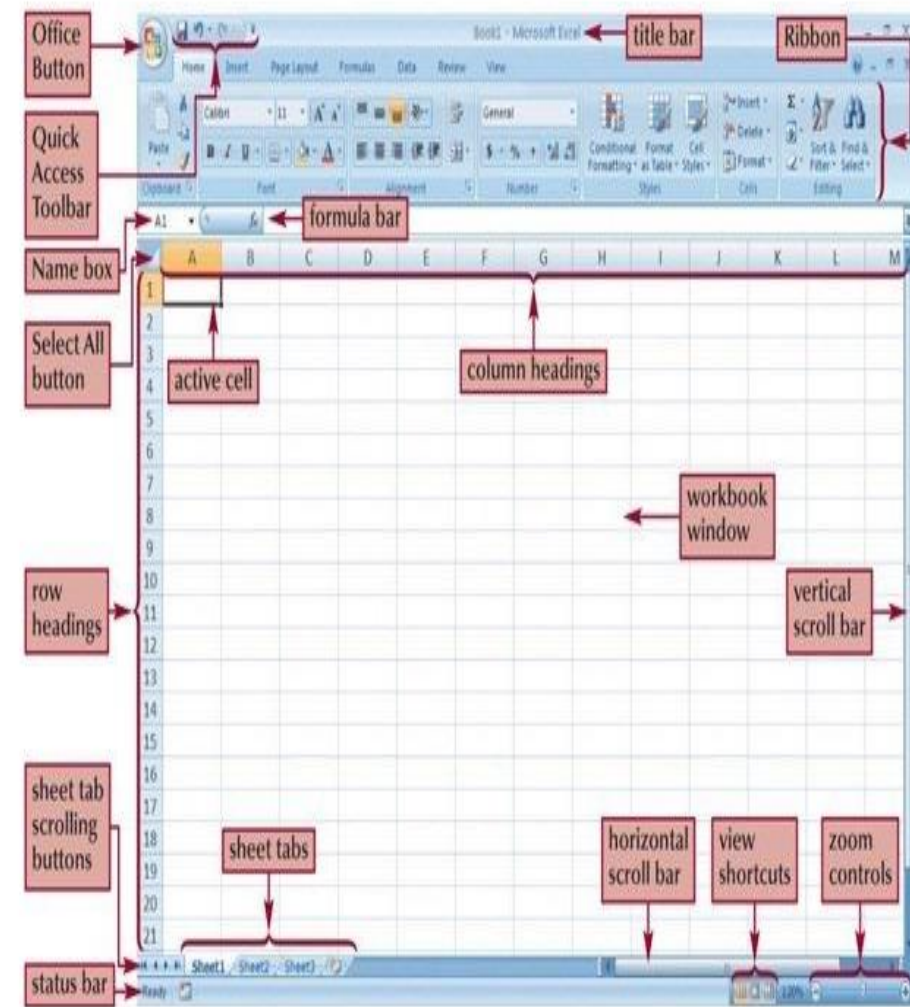
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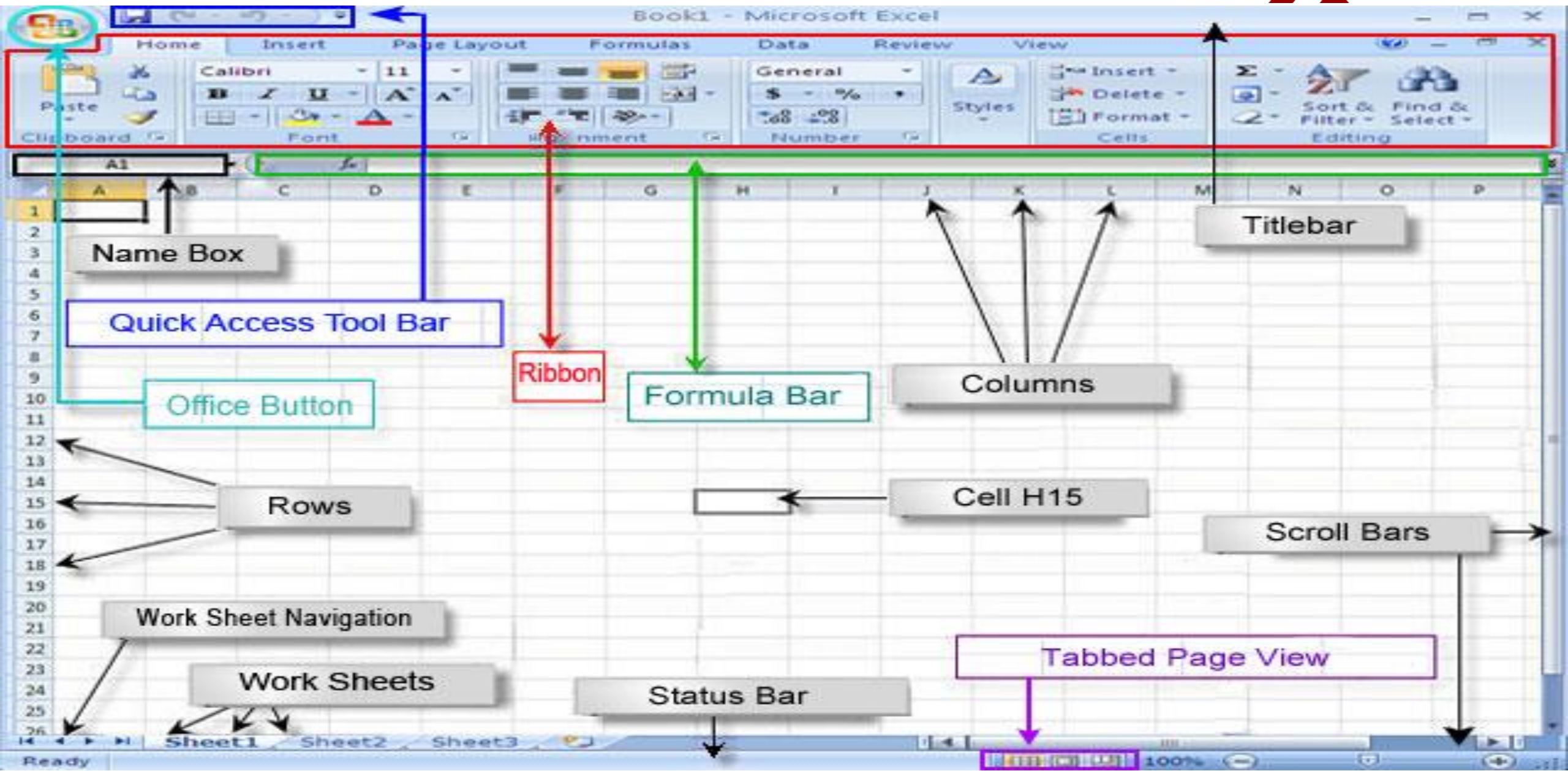
Worked with top service and product companies like Wipro , Volvo , TVS , TEG Analytics, General Electrics , Royal Bank of Scotland , Verizon etc.

Worked with Universities like Praxis, IIDT, Snt. Claret, Matrushi, Jain University , RGTU

- Introduction to Excel 2007, Introduction to data, Cell address, ExcelData Types,
- Concept of hyperlink, Introduction to formatting, number,text and date formatting,
- Concept of worksheet and workbook,Understanding formulas, Operators in Excel 2007,
- Operators Precedence, Understanding Functions, Common Excel Functions suchas sum, average, min, max, date, transpose, In, And, or, sqrt, power,upper, lower.
- Types of graphics : Word art, auto shapes , Images.
- Introduction to charts: overview of different types of charts available with Excel
- Concept of print area, margins, header, footer and other page setup options

- **MS-EXCEL is a part of Microsoft Office suite software.** It is an electronic spreadsheet with numerous rows and columns, used for organizing data, graphically represent data(s), and performing different calculations.
- It consists of 1048576 rows and 16383 columns, a row and column together make a cell.
- Excel is a **spreadsheet program from Microsoft and a component of its Office product group for business applications.**
- Microsoft Excel enables users to format, organize and calculate data in a spreadsheet.
- Excel data types are the four different kinds of values in Microsoft Excel. The four types of data are
- **text, number, logical and error.**

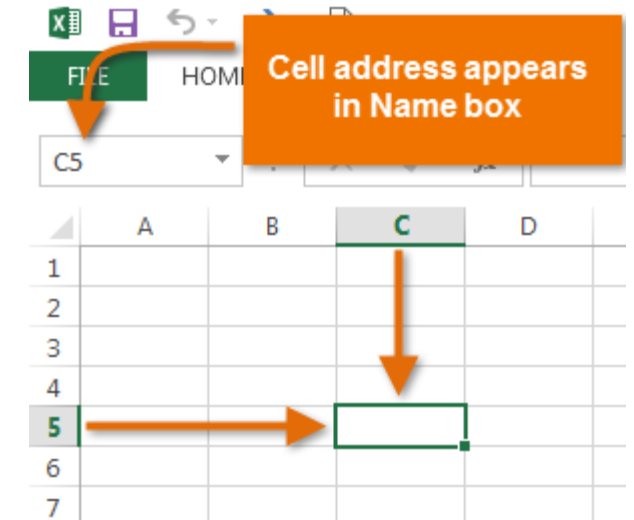




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- A cell reference or cell address is a combination of a column letter and a row number that identifies a cell on a worksheet.
- For example, A1 refers to the cell at the intersection of column A and row 1; B2 refers to the second cell in column B, and so on.
- A reference is a cell's address. It identifies a cell or range of cells by referring to the column letter and row number of the cell(s).



Excel data types are the four different kinds of values in Microsoft Excel. The four types of data are

4 Excel data types

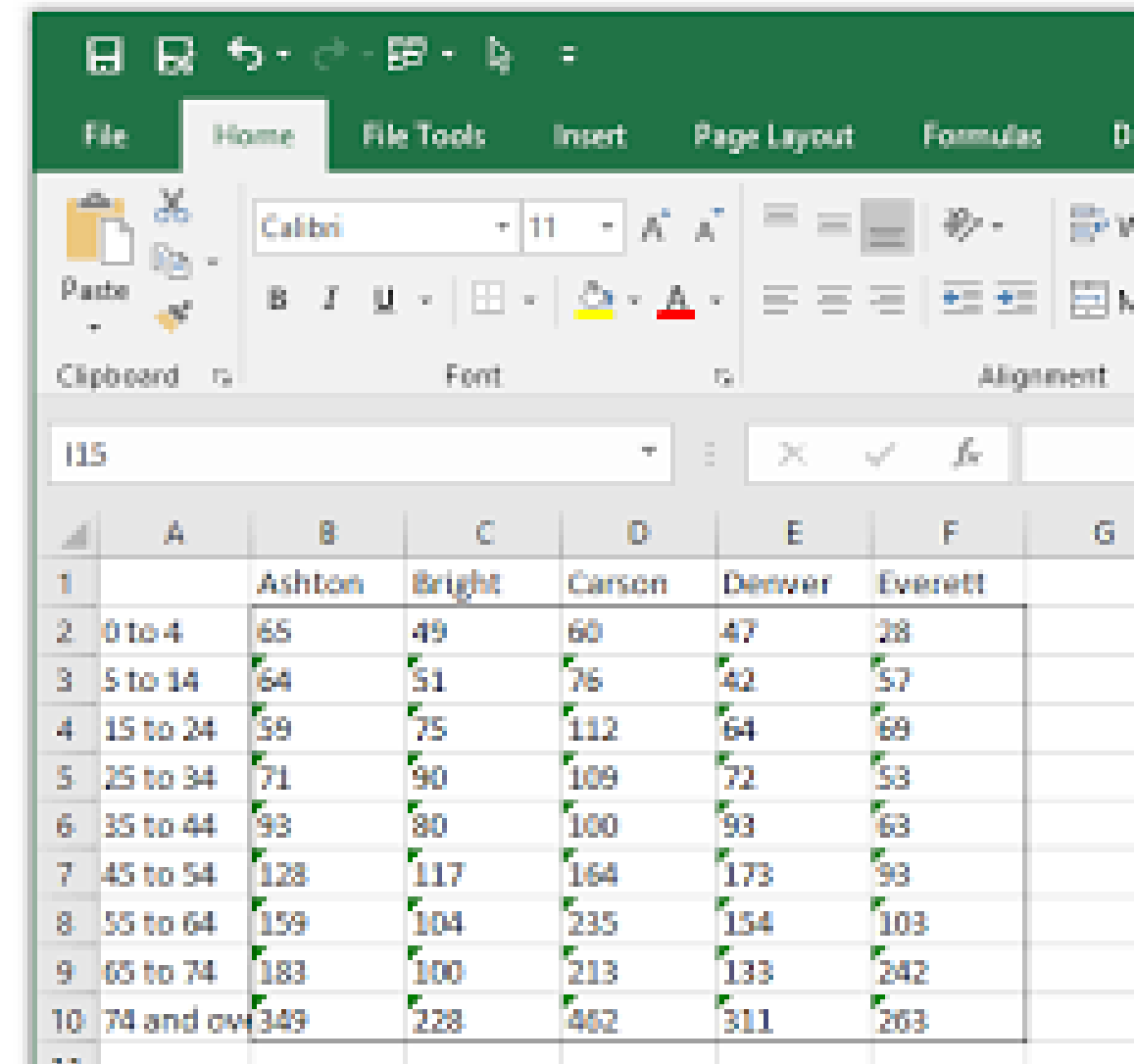
1. Number data
2. Text data
3. Logical data
4. Error data



- Data in this category includes any kind of number. These may include large numbers or small fractions and quantitative or qualitative data.
- Examples of number data may include:
 - Monetary totals
 - Whole numbers
 - Percentages
 - Decimals
 - Dates
 - Times
 - Integers
 - Phone numbers

Available number formats in Excel

1. Select a cell or a cell range.
2. On the Home tab, select Number from the drop-down.
Or, you can choose one of these options: Press CTRL + 1 and select Number. Right-click the cell or cell range, select Format Cells... , and select Number. ...
3. Select the format you want.

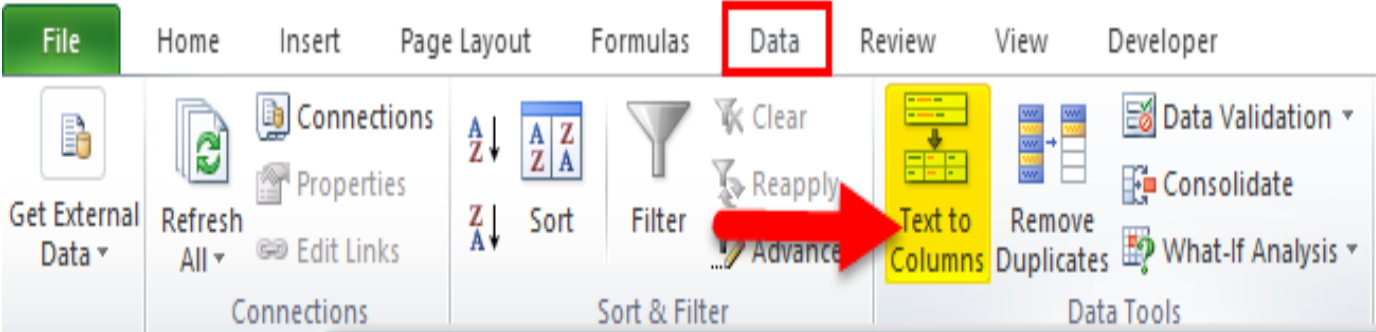


- This kind of data includes characters such as alphabetical, numerical and special symbols. The primary difference between number data and text data is that you can use calculations on number data but not text data.
- Since there can be overlap between these two types of data, you may manually change the format of a cell to ensure it operates the way you want.
- Excel may categorize figures it doesn't recognize as text data by default, so it's important to format your data to fit the type you want. Examples of text data may include:
 - Words
 - Sentences
 - Dates
 - Times
 - Addresses

Text to Columns

- 1.Select the range with full names.
- 2.On the Data tab, in the Data Tools group, click Text to Columns. The following dialog box appears.
- 3.Choose Delimited and click Next.
- 4.Clear all the check boxes under Delimiters except for the Comma and Space check box.
- 5.Click Finish.

Text to Columns in Excel



H20

A

1	Timestamp
2	11/1/2018 9:52
3	11/1/2018 10:0
4	11/1/2018 10:0
5	11/1/2018 10:0
6	11/1/2018 10:0
7	11/1/2018 10:0
8	11/1/2018 10:0

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

☒ Delimited

- Characters such as commas or tabs separate each field.

☐ Fixed width

- Fields are aligned in columns with spaces between each field.

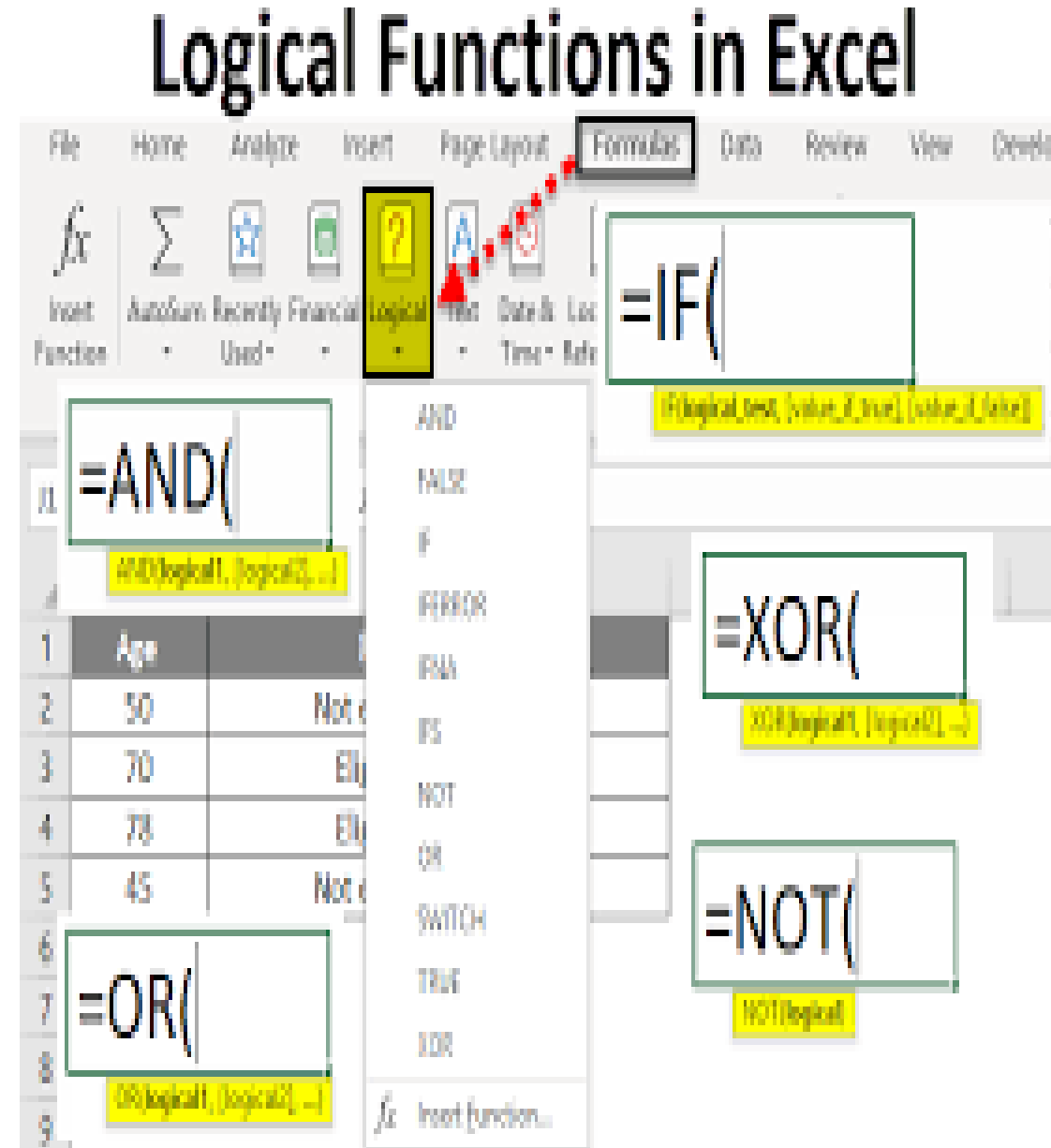
Preview of selected data:

1 Last Name

3.Logical data

- Data in this type is either TRUE or FALSE, usually as the product of a test or comparison. This means you can use a function to determine whether the data in your spreadsheet meets different measures.
- Microsoft Excel provides 4 logical functions are
- **AND,**
- **OR,**
- **XOR and**
- **NOT.**

Logical Functions in Excel



The screenshot displays the Microsoft Excel interface with the 'Formulas' tab selected. The 'Logical' group in the ribbon is highlighted, and a dropdown menu is open, listing various logical functions: AND, FALSE, IF, IFERROR, IFNA, IS, NOT, OR, SWITCH, TRUE, and XOR. Below the menu, four function boxes are shown, each with a yellow tooltip indicating its syntax:

- =AND()**: AND(logical1, [logical2], ...)
- =OR()**: OR(logical1, [logical2], ...)
- =XOR()**: XOR(logical1, [logical2], ...)
- =NOT()**: NOT(logical)

- **AND:** An AND function may help you determine whether your data meets multiple conditions. For example, you might use this function to test if data in one cell is larger than a certain amount and the data in another cell is also larger than another amount.
- The AND function **returns TRUE if all its arguments evaluate to TRUE, and returns FALSE if one or more arguments evaluate to FALSE**. One common use for the AND function is to expand the usefulness of other functions that perform logical tests.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				

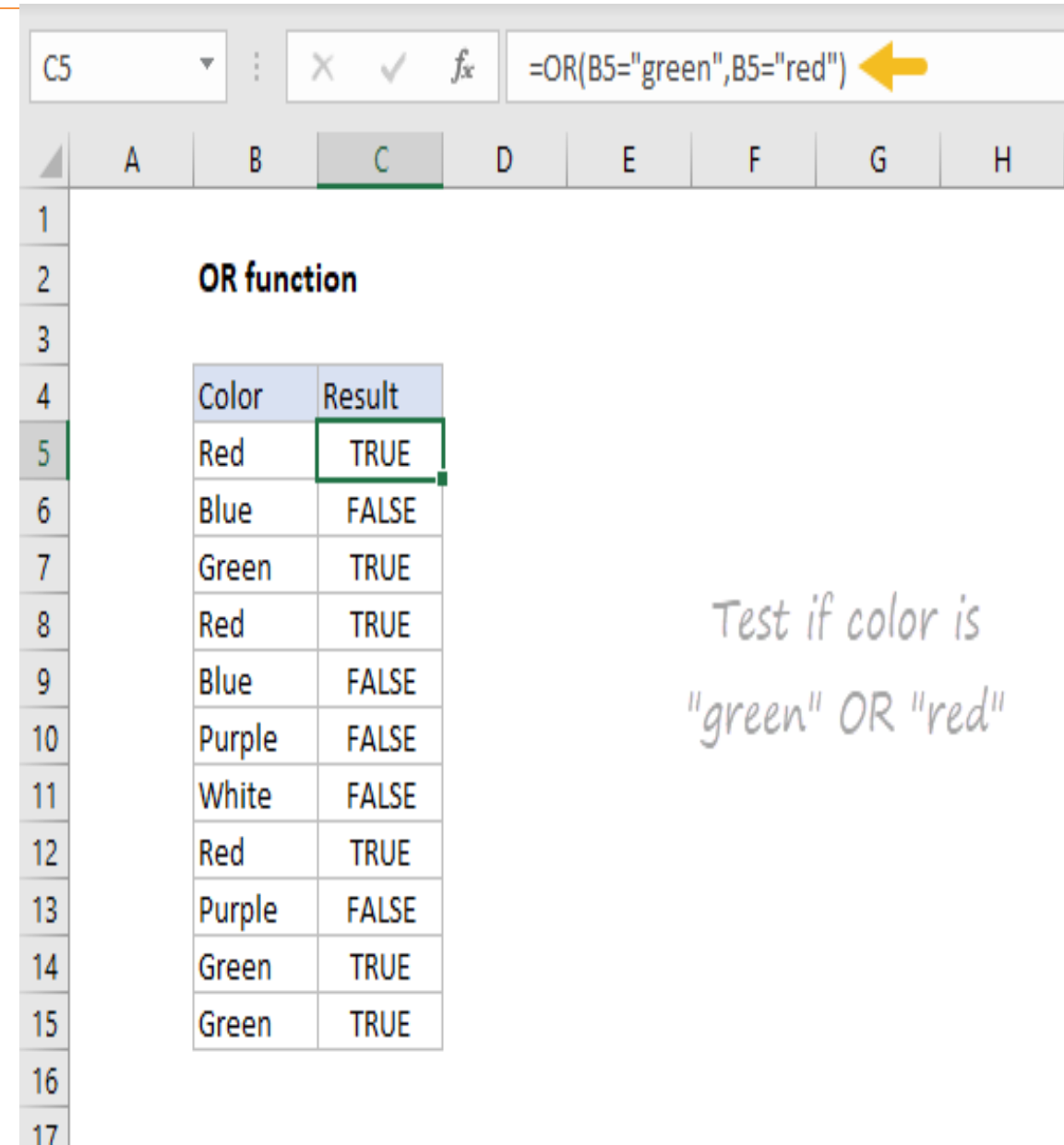
AND Function Overview

Number	Less Than 80, Greater Than 40
52	TRUE
22	FALSE
79	TRUE
41	TRUE
42	TRUE
94	FALSE
81	FALSE
67	TRUE
76	TRUE
50	TRUE
82	FALSE
72	TRUE
68	TRUE

>>> =AND(B5>40, B5<80)
logic1: B5>40
logic 2: B5<80

OR FUNCTION

- You may use this function to determine that at least one of your arguments meets your conditions. If none of the data matches your conditions, Excel produces a FALSE value.
- The OR function **returns TRUE if any of its arguments evaluate to TRUE, and returns FALSE if all of its arguments evaluate to FALSE**. One common use for the OR function is to expand the usefulness of other functions that perform logical tests.



The screenshot shows an Excel spreadsheet with the formula bar at the top displaying `=OR(B5="green",B5="red")`. A yellow arrow points to the formula bar. Below the formula bar, the spreadsheet grid shows columns A through H and rows 1 through 17. In cell C2, the text "OR function" is displayed. Below this, a table with two columns, "Color" and "Result", is shown. The table contains the following data:

Color	Result
Red	TRUE
Blue	FALSE
Green	TRUE
Red	TRUE
Blue	FALSE
Purple	FALSE
White	FALSE
Red	TRUE
Purple	FALSE
Green	TRUE
Green	TRUE

To the right of the table, there is a handwritten note in grey text: "Test if color is 'green' OR 'red'" data-bbox="820 500 960 620"/>

- XOR with more logicals**

Returns TRUE only when number of TRUEs is odd

L1	L2	L3	L4	L5	Result
0	0	0	0	0	FALSE
1	0	0	0	0	TRUE
1	1	0	0	0	FALSE
1	1	1	0	0	TRUE
1	1	1	1	0	FALSE
1	1	1	1	1	TRUE

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NOT FUNCTION

- You might use this function when you want to filter out arguments that don't match your conditions. This marks each argument as TRUE so you can assess possible patterns in data that doesn't match your conditions.
- By using the NOT function as the logical_test argument of the IF function, you can test many different conditions instead of just one. The NOT function syntax has the following arguments: **Logical Required. A value or expression that can be evaluated to TRUE or FALSE.**

C2 : ✕ ✓ fx =NOT(B2="Banana")

	A	B	C
1	Item	Product	Any product but Banana
2	20191101	Apple	TRUE
3	20191102	Lemon	TRUE
4	20191103	Grape	TRUE
5	20191104	Banana	FALSE
6	20191105	Cherry	TRUE
7	20191106	Watermelon	TRUE
8	20191107	Banana	FALSE

D6 : ✕ ✓ fx =NOT(B6>30)

	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

NOT function
Reverses a logical or boolean result value

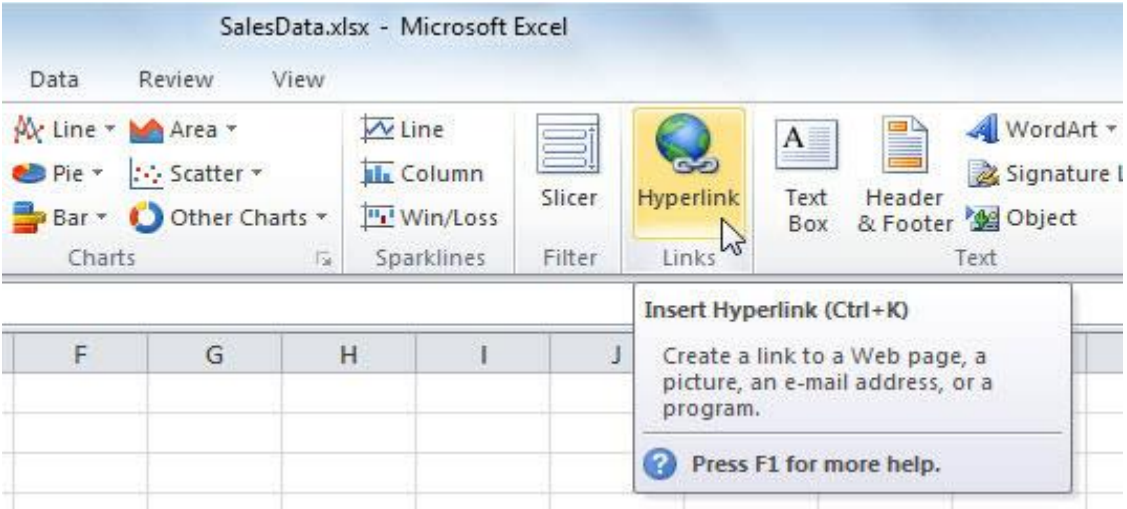
Age	> 30	NOT(> 30)
31	TRUE	FALSE
23	FALSE	TRUE
19	FALSE	TRUE
40	TRUE	FALSE
36	TRUE	FALSE
51	TRUE	FALSE

- The **HYPERLINK** function creates a shortcut that jumps to another location in the current workbook, or opens a document stored on a network server, an intranet, or the Internet. When you click a cell that contains a **HYPERLINK** function, Excel jumps to the location listed, or opens the document you specified.

- Syntax

HYPERLINK(link_location, [friendly_name])

Excel has a **HYPERLINK** function that you can also use for creating links between spreadsheets in the workbook.



D5

✕

✓

f_x

=HYPERLINK(C5,B5)

A

B

C

D

E

F

G

1

2

3

4

5

6

7

8

9

10

11

12

HYPERLINK function

Name	Target	Hyperlink
Exceljet	https://exceljet.net	Exceljet
Google	https://www.google.com	Google
Sheet2	#Sheet2!A1	Sheet2
Total amount	#Z100	Total amount
Summary	#summary	Summary
another workbook	another workbook.xlsx	another workbook
email me	mailto: bill@acme.com	email me

// cell A1 in Sheet2

// cell Z100 in same sheet

// named range "summary"

// workbook in same folder

// an email link

File Home Insert Page Layout Formulas Data Review View

PivotTable Table Picture Clip Art Charts Sparklines Hyperlink Text Box Header & Footer Symbols

Insert Hyperlink

Article 1_Hyperlinks.xlsm - Module1 (Code)

Link to: Text to display: C:\Users\xyz_1\Desktop\hyperlink\Article 1 Hyperlinks.xlsm

(General)

Option Explicit

Sub Create_Hyperlink()

Dim Ws As Worksheet

Dim LR As Long

Dim Answer As VbMsgBoxResult

Answer = MsgBox("Do you want to create a hyperlink?", vbYesNo)

For Each Ws In Worksheets

Ws.Visible = True

Ws.Hyperlinks.Add Anchor:=Nothing, Address:=C:\Users\xyz_1\Desktop\hyperlink\Article 1 Hyperlinks.xlsm, TextToDisplay:=C:\Users\xyz_1\Desktop\hyperlink\Article 1 Hyperlinks.xlsm

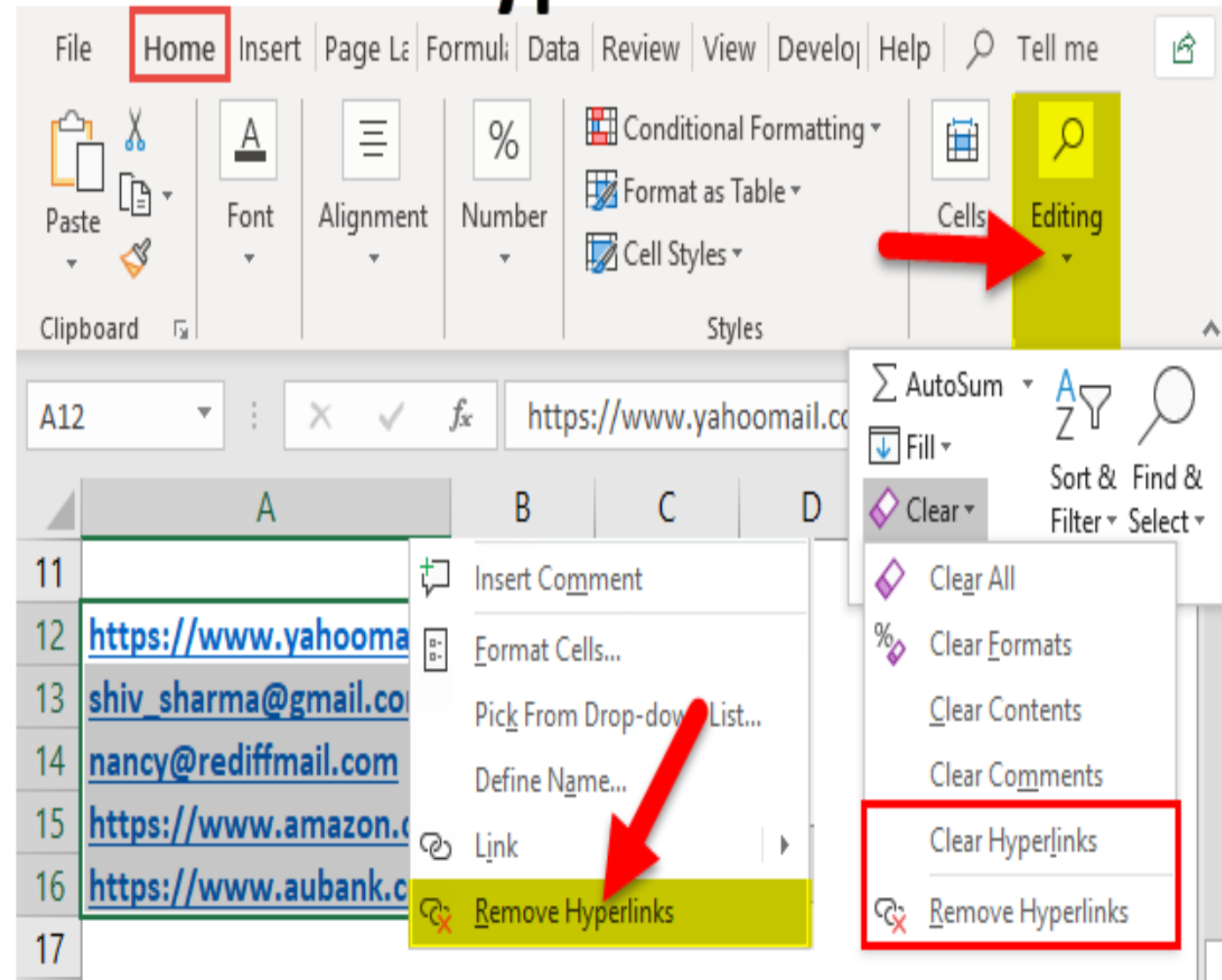
Next Ws

End Sub

1. On a worksheet, select the cell where you want to create a link.
2. On the Insert tab, select Hyperlink. ...
3. Under Display Text:, type the text that you want to use to represent the link.
4. Under URL:, type the complete Uniform Resource Locator (URL) of the webpage you want to link to.
5. Select OK.

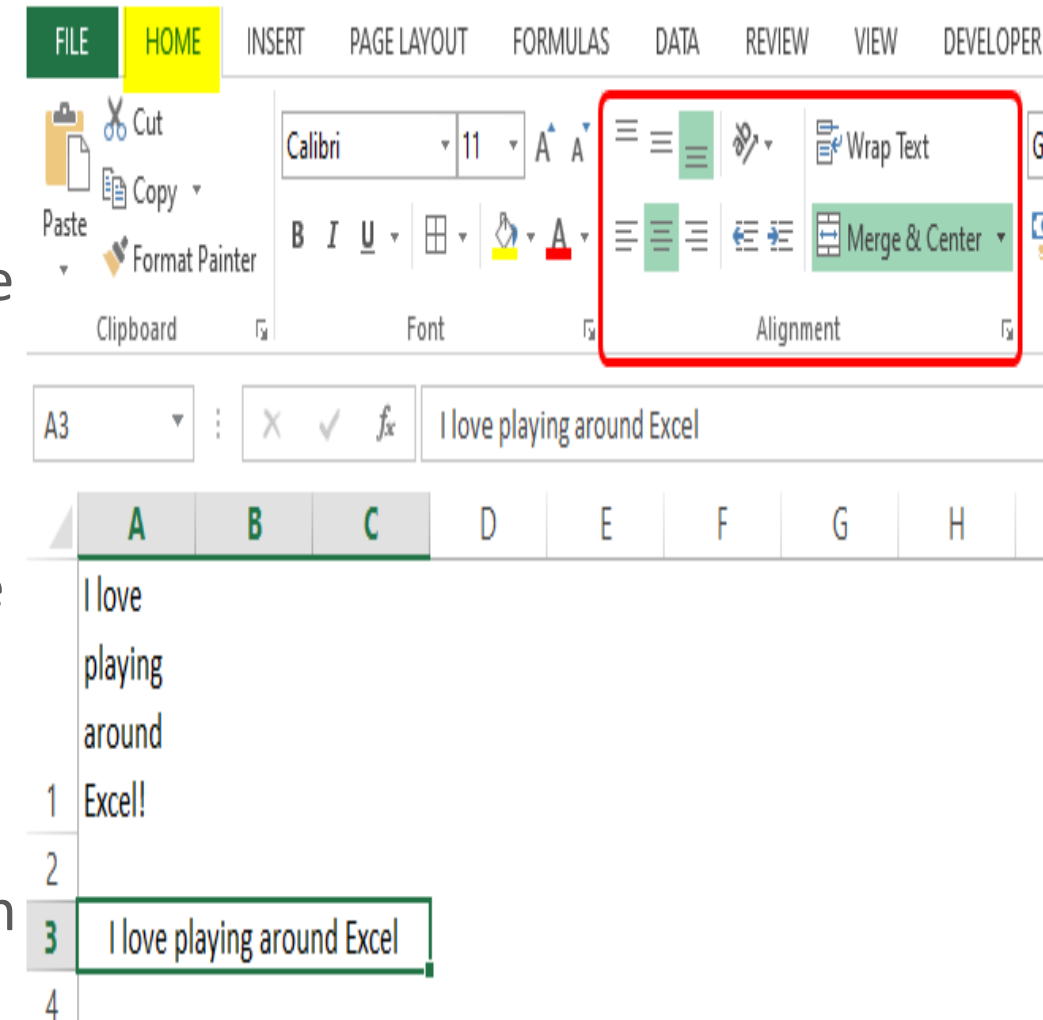
Select all cells that contain hyperlinks, or press Ctrl+A to select all cells. **Right-click, and then click Remove Hyperlinks.**

Remove Hyperlinks in Excel



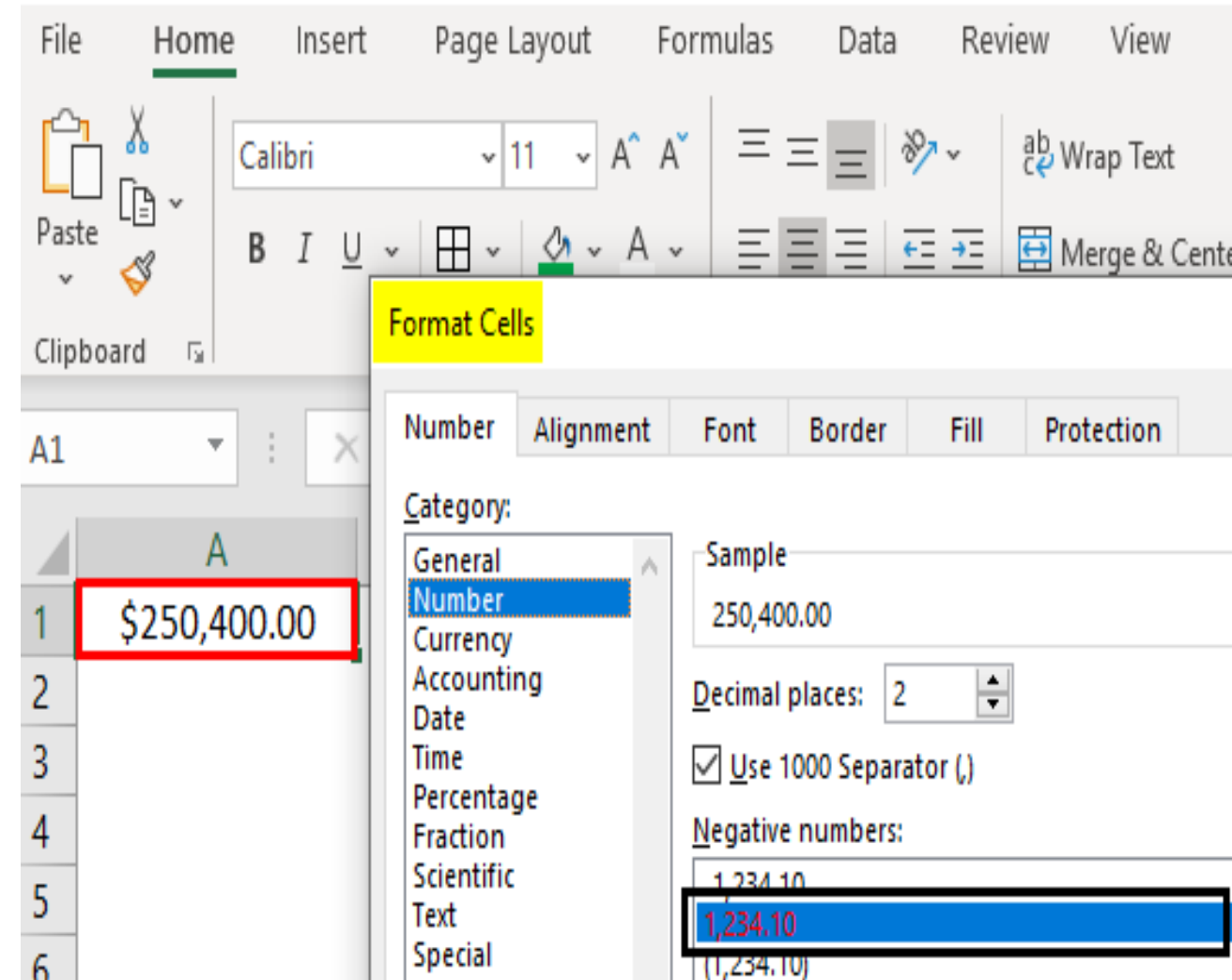
- Formatting is **the process of turning the result of an XSL transformation into a tangible form for the reader or listener**. This process comprises several steps, some of which depend on others in a non-sequential way.
- Formatting also makes information more accessible to the reader by creating and labeling sections (headings), highlighting key words or ideas (bold, italics, or lists), and making a good impression (professional look and feel, appropriate font choice for the document type).
- **Formatting in Excel** means a trick that we can use to modify the data's appearance in a worksheet. We can format the data in various ways, like we can format

Formatting Text in Excel

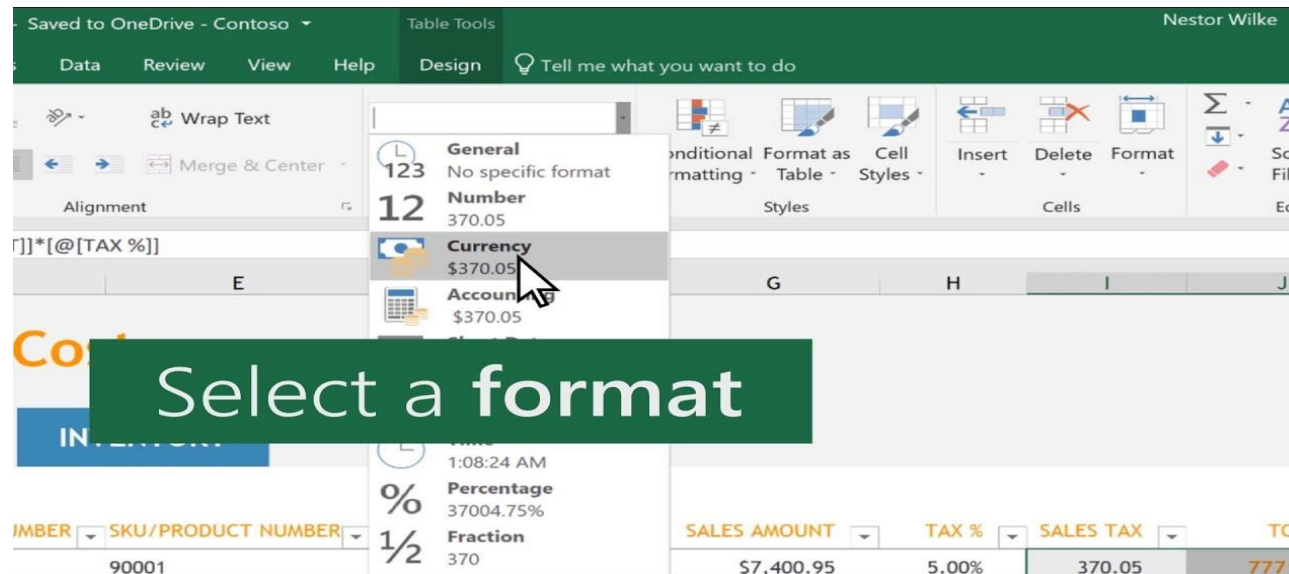
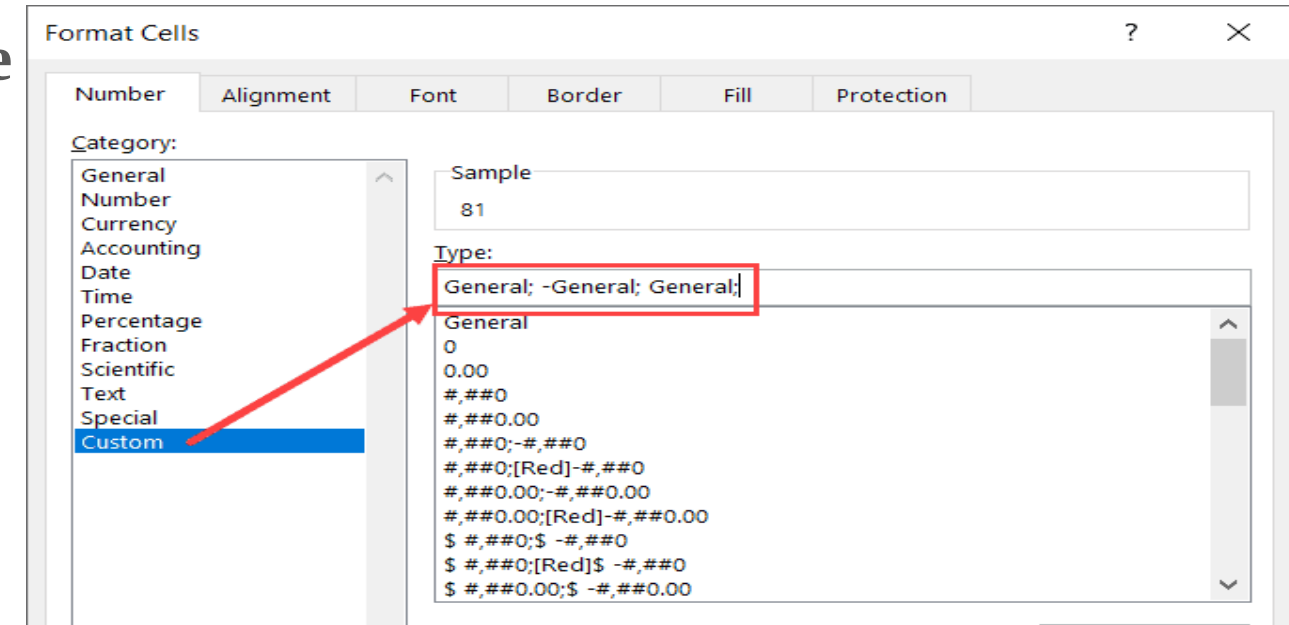


- Select the cells with the Excel Style. **Right-click the applied style in Home > Cell Styles. Select Modify > Format to change what you want.**
- Cell formats **allow you to only change the way cell data appears in the spreadsheet.**
- It is important to keep in mind that it only alters the way the data is presented, and does not change the value of the data.
- The formatting options allows for monetary units, scientific options, dates, times, fractions, and more.

Format Cells in Excel



- You can use number formats to **change the appearance of numbers, including dates and times, without changing the actual number.**
- The number format does not affect the cell value that Excel uses to perform calculations. The actual value is displayed in the formula bar.
- Excel provides several built-in number formats. You can select standard number formats (**General, Number, Currency, Accounting, Short Date, Long Date, Time, Percentage, Fraction, Scientific, Text**) on the home tab of the ribbon using the Number Format menu.



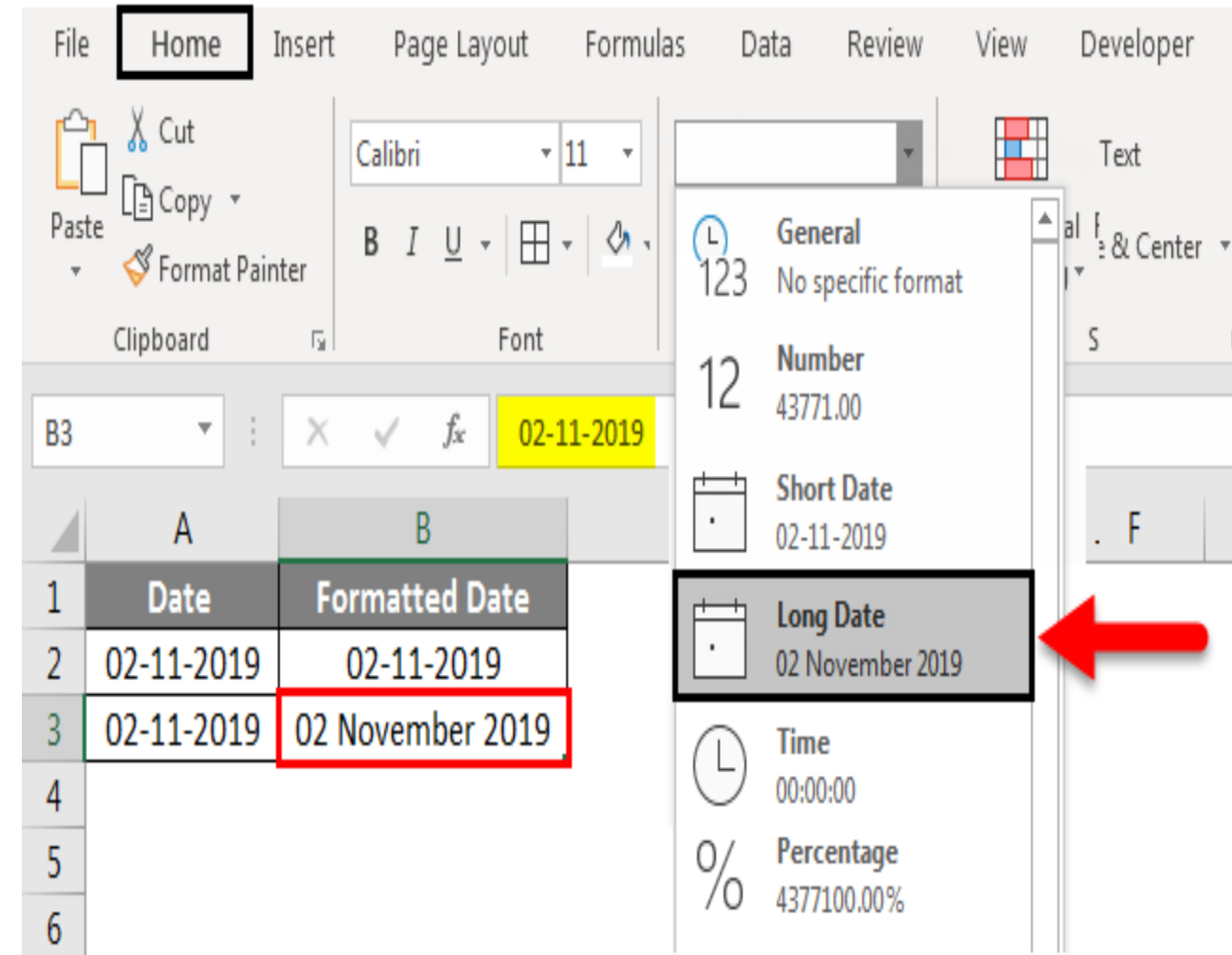
- Formatting text or numbers can make them appear more visible especially when you have a large worksheet. Changing default formats includes things like **changing the font color, style, size**
- **You can use the Format Cells dialog to find the other available format codes:**
- Press Ctrl+1 (+1 on the Mac) to bring up the Format Cells dialog.
- Select the format you want from the Number tab.
- Select the Custom option,
- The format code you want is now shown in the Type box.

Formatting Text in Excel

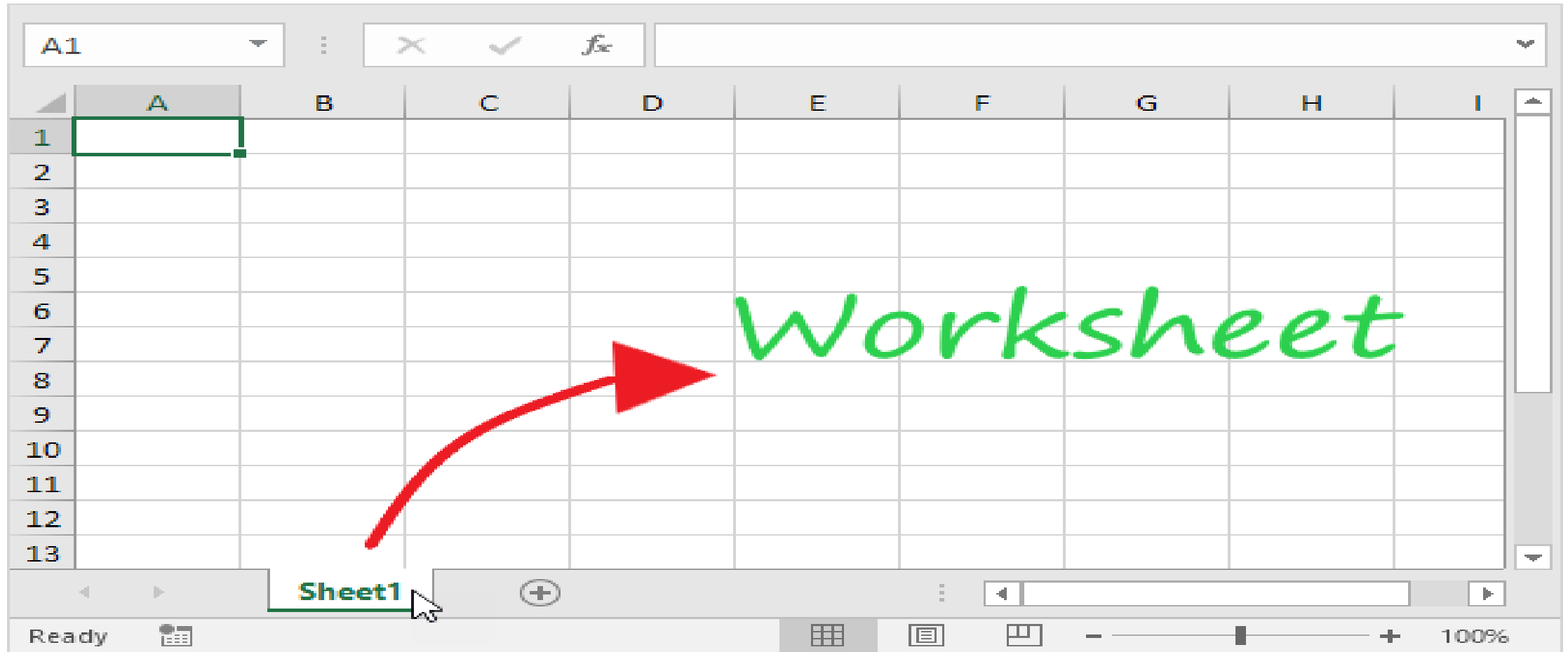
The screenshot illustrates the Excel interface for text formatting. The **Home** tab is active, with the **Font** and **Alignment** groups highlighted. The **Font** group shows the font set to **Arial**, size **20**, with **B** (bold), **I** (italic), and **U** (underline) options. The **Alignment** group shows text alignment options. Below the ribbon, the worksheet grid shows cell **A1** containing the text **Hello Excel** in red, bold, italicized font. To the right, the **Conditional Formatting** dropdown menu is open, showing options like **Highlight Cells Rules**, **Top/Bottom**, **Equal To...**, **Text that Contains...**, **Data**, **Color Scales**, **Duplicate Values...**, and **More Rules...**.

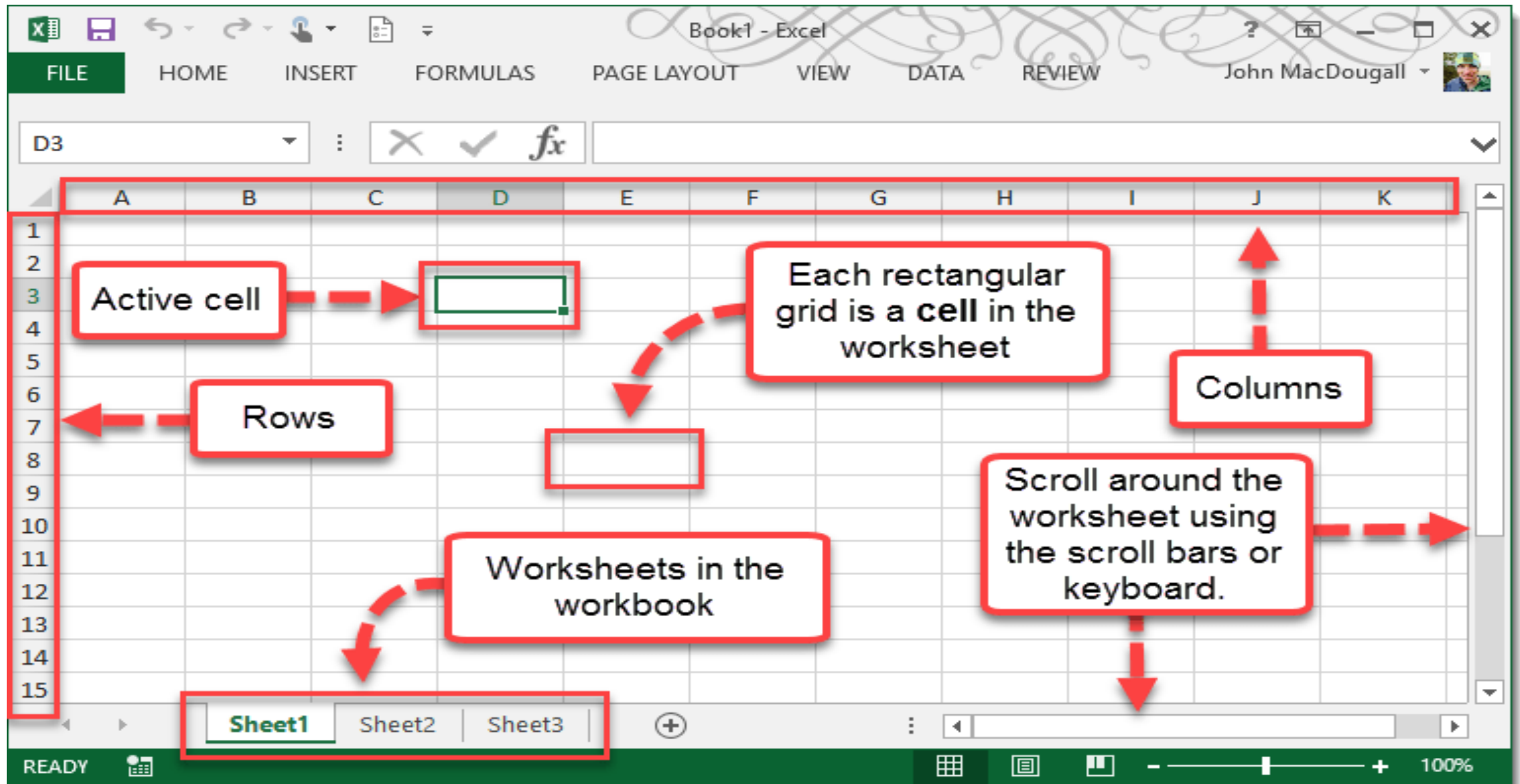
- Select the cells you want to format.
- Press CTRL+1.
- In the Format Cells box, click the Number tab.
- In the Category list, click Date.
- Under Type, pick a date format. ...
- If you want to use a date format according to how another language displays dates, choose the language in Locale (location).
- First, pick the cells that contain dates, then right-click and select Format Cells. Select Custom in the Number Tab, then type 'dd-mmm-yyyy' in the Type text box, then click okay. It will format the dates you specify.

Date Format in Excel



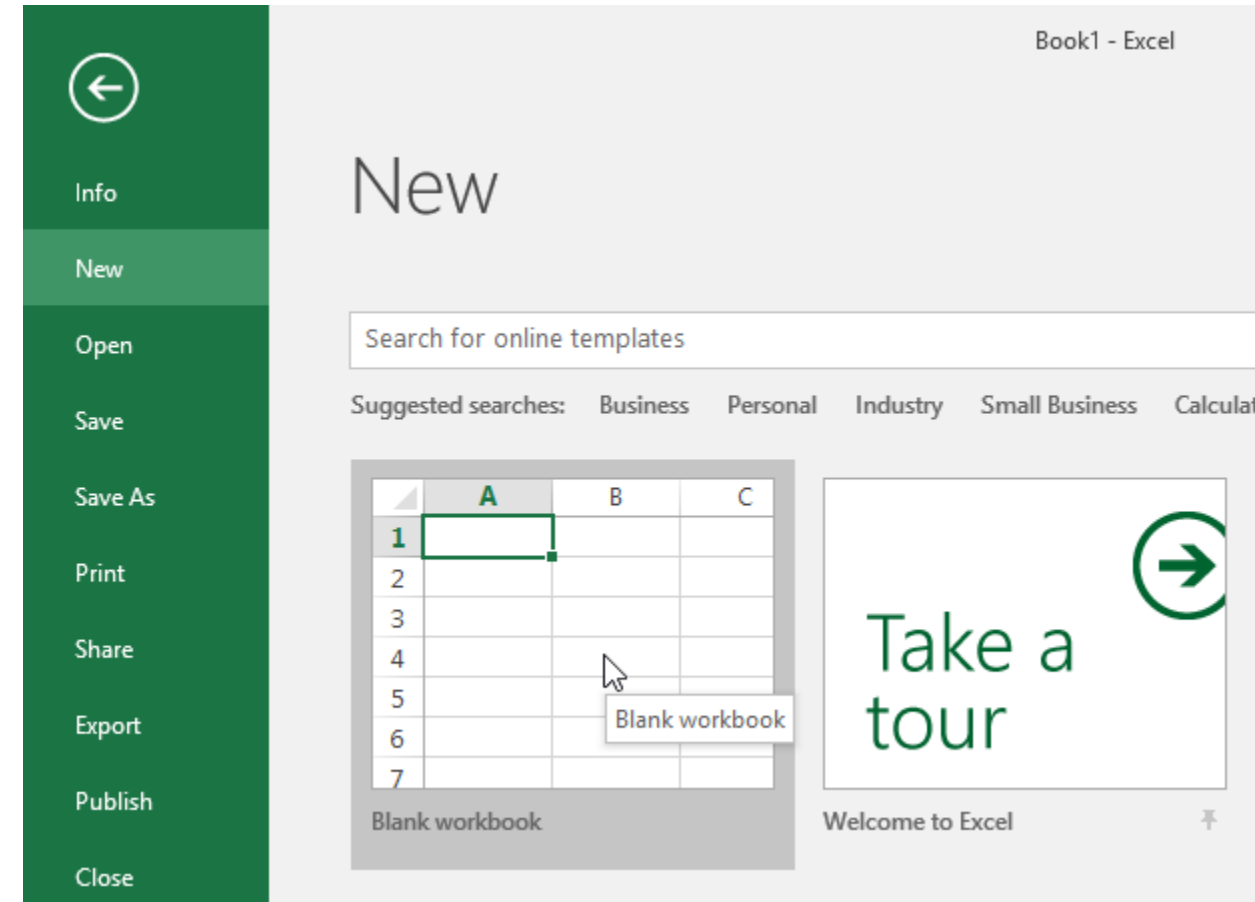
- A Worksheet is basically a single-page spreadsheet containing information. A workbook is a file that contains multiple spreadsheets. A worksheet contains a matrix of rectangular cells, organized in a form of rows and columns.

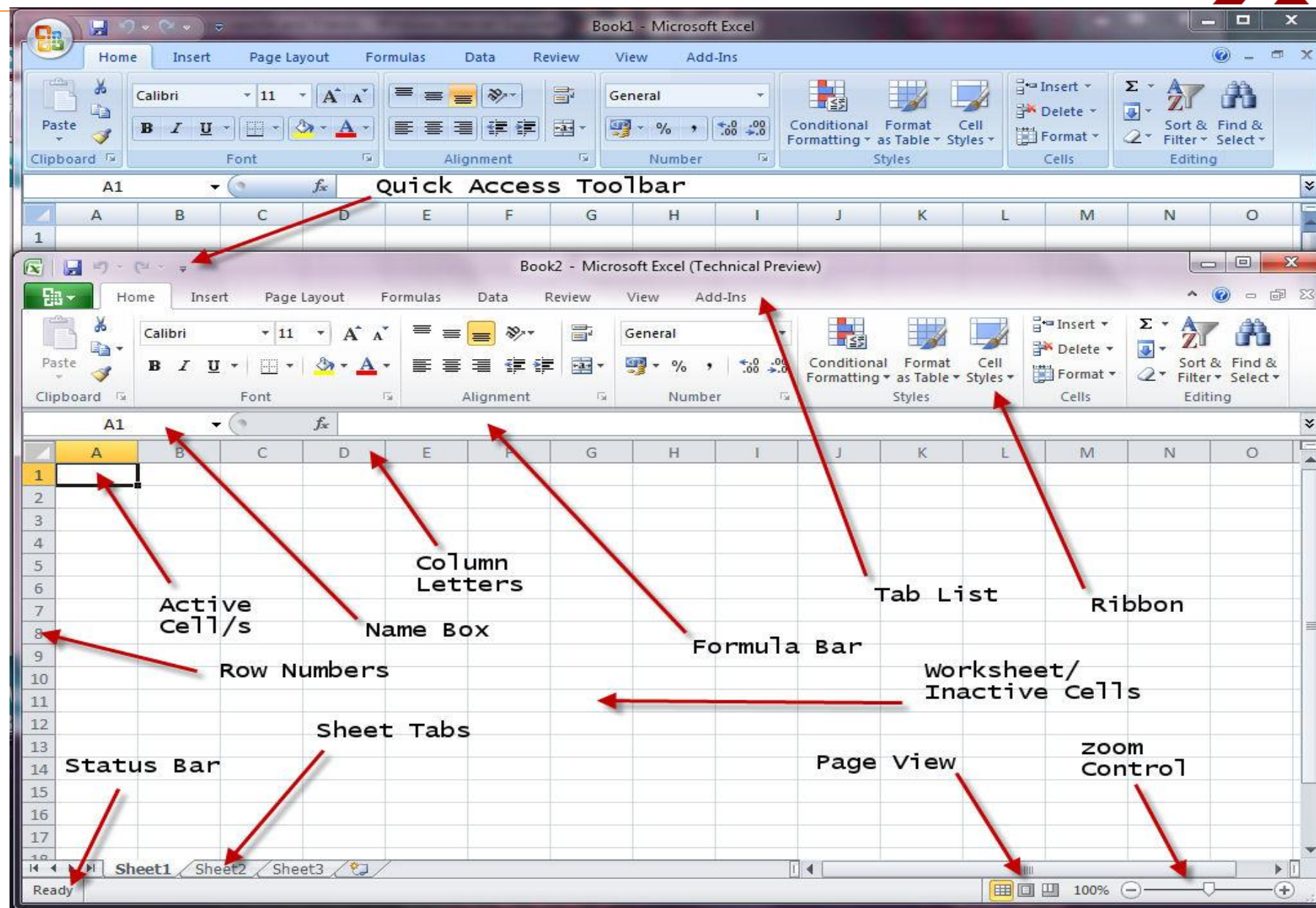




- **A workbook is a file that contains multiple spreadsheets.** A worksheet contains a matrix of rectangular cells, organized in a form of rows and columns. A workbook contains one or more worksheets, consisting of related information.

Click the File tab. Click New. Under Available Templates, double-click Blank Workbook. Keyboard shortcut To quickly create a new, blank workbook, you can also press CTRL+N.

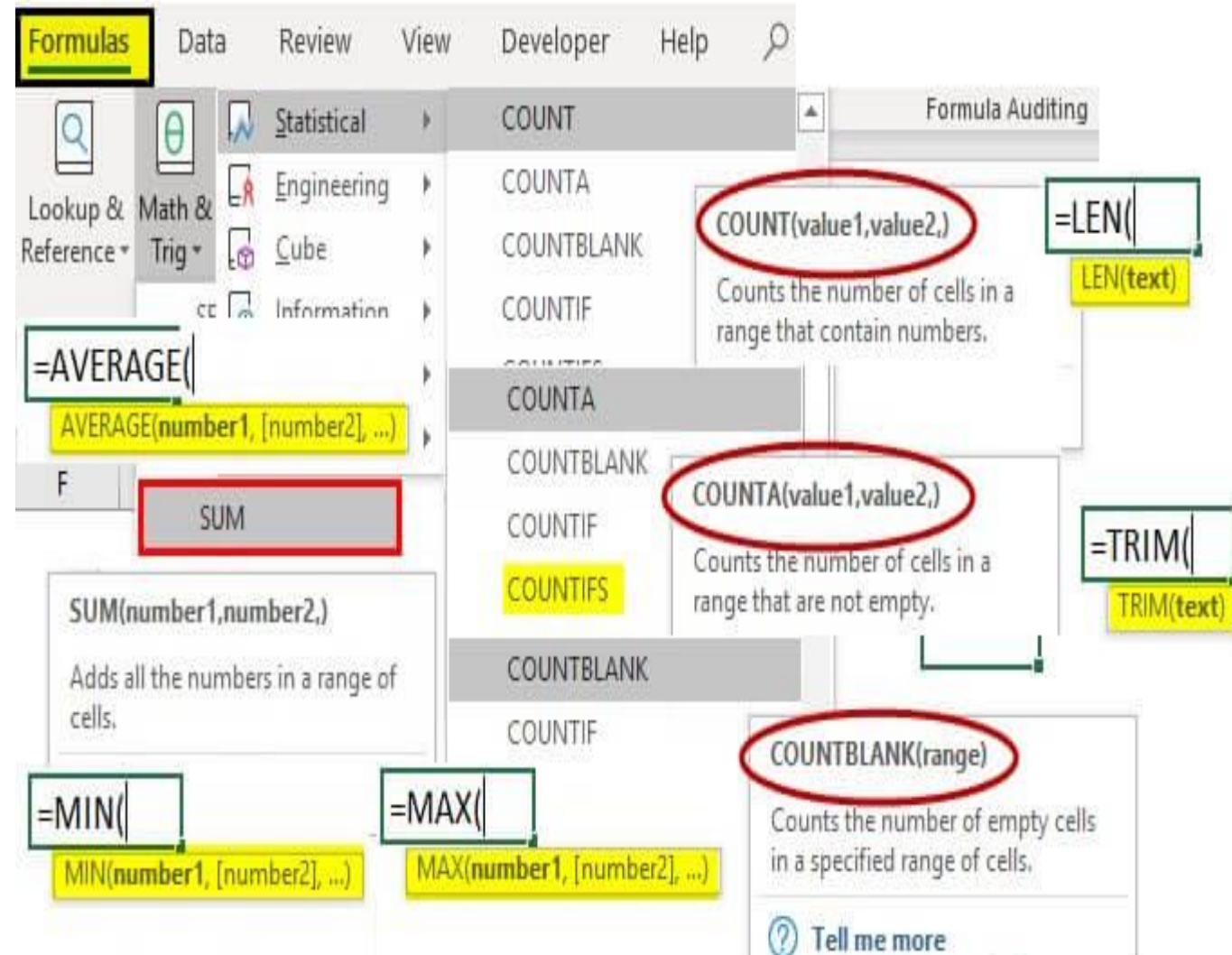




Enter a formula that contains a built-in function

- Select an empty cell.
- Type an equal sign = and then type a function. For example, =SUM for getting the total sales.
- Type an opening parenthesis (.
- Select the range of cells, and then type a closing parenthesis).
- Press Enter to get the result.

Basic Excel Formulas



The screenshot displays the 'Formulas' ribbon in Microsoft Excel, showing a list of functions categorized by type. The 'Statistical' category is expanded, listing functions like COUNT, COUNTA, COUNTBLANK, COUNTIF, COUNTIFS, and COUNTBLANK. The 'Formula Auditing' tab is also visible. Several functions are highlighted with red circles and their syntax is shown in a separate box:

- COUNT(value1,value2,)**: Counts the number of cells in a range that contain numbers.
- COUNTA(value1,value2,)**: Counts the number of cells in a range that are not empty.
- COUNTBLANK(range)**: Counts the number of empty cells in a specified range of cells.
- =LEN(text)**: Returns the number of characters in a text string.
- =TRIM(text)**: Removes extra spaces from a text string.
- =AVERAGE(AVERAGE(number1, [number2], ...))**: Returns the average of a range of numbers.
- SUM(SUM(number1,number2,))**: Adds all the numbers in a range of cells.
- =MIN(MIN(number1, [number2], ...))**: Returns the minimum value in a range of numbers.
- =MAX(MAX(number1, [number2], ...))**: Returns the maximum value in a range of numbers.

Here is the list of the top 10 basic formulas and functions in Excel.

- 1.SUM
- 2.COUNT
- 3.AVERAGE
- 4.MIN Excel
- 5.MAX Excel
- 6.LEN Excel
- 7.RAND Excel
- 8.IF Excel
- 9.UPPER
- 10.LOWER
- 11.NOW
- 12.DATE
- 13.PROPER

Basic Formulas in Excel

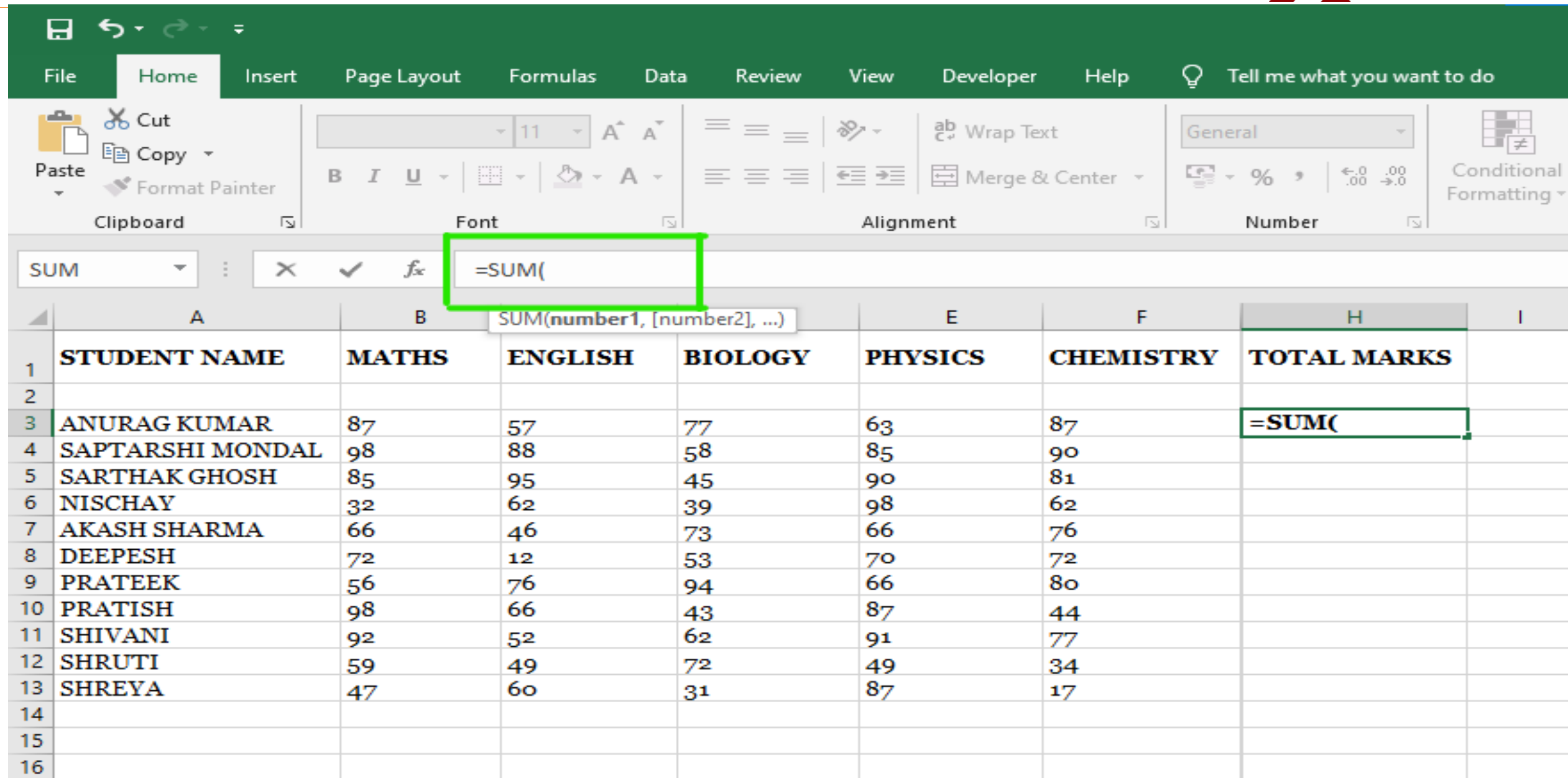
The screenshot shows the Microsoft Excel interface with the 'Formulas' tab selected. The 'Function Library' group is highlighted with a red box, containing icons for Insert Function, AutoSum, Recently Used, Financial, Logical, Text, Date & Time, Lookup & Reference, Math & Trig, and More Functions. Below the ribbon, a grid displays 13 basic formulas, each with its syntax in a yellow box:

=SUM(SUM(number1, [number2], ...)	=MAX(MAX(number1, [number2], ...)	=MIN(MIN(number1, [number2], ...)	=NOW(NOW()	
=AVERAGE(AVERAGE(number1, [number2], ...)	=COUNT(COUNT(value1, [value2], ...)	=COUNTA(COUNTA(value1, [value2], ...)	=LEN(LEN(text)	
=ABS(ABS(number)	=RAND(RAND()	=RANDBETWEEN(RANDBETWEEN(bottom, top)	=UPPER(UPPER(text)	=LOWER(LOWER(text)
		=PROPER(PROPER(text)		

- The SUM function adds values. You can add individual values, cell references or ranges or a mix of all three.

SUM Function in Excel

The screenshot shows the Microsoft Excel ribbon with the 'Formulas' tab selected. The 'Math & Trig' group is highlighted with a red arrow. The 'SUM' function is selected from the dropdown menu, also highlighted with a red arrow. A tooltip for the SUM function is displayed, showing the formula `SUM(number1,number2,...)` and the description 'Adds all the numbers in a range of cells.' A red box highlights the tooltip. In the background, the Excel grid shows the formula bar with `=SUM(` and a yellow tooltip for the SUM function syntax.

ALGAE

- The AVERAGE function in Excel **calculates the arithmetic mean of the supplied values**. Such values can be numbers, percentages or times. Returns the average (arithmetic mean) of the arguments. For example, if the range A1:A20 contains numbers, the formula **=AVERAGE(A1:A20)** returns the average of those numbers.

AVERAGE in Excel

=AVERAGE(

AVERAGE(number1, [number2], ...)

AVERAGE(number1,number2,)

Returns the average (arithmetic mean) of its arguments, which can be numbers or names, arrays, or references that contain numbers.

Press F1 for more help.

Average Formula in Excel

AVERAGE

=AVERAGE(B2:B11)

	A	B	C	D	E	F
1	Date of Delivery	Quantity		Average Formula	33.6	
2	19/2/2019	55				
3	20/2/2019	12				
4	21/2/2019	21				
5	22/2/2019	12				
6	23/2/2019	23				
7	24/2/2019	55				
8	25/2/2019	67				
9	26/2/2019	1				
10	27/2/2019	34				
11	29/2/2019	56				

Search for a function:

AVERAGE

Or select a category: **Statistical**

Select a function:

AVERAGE

AVERAGE(number1, [number2], ...)

ALGAE

AVERAGE **=AVERAGE (number1, [number2], ...)**

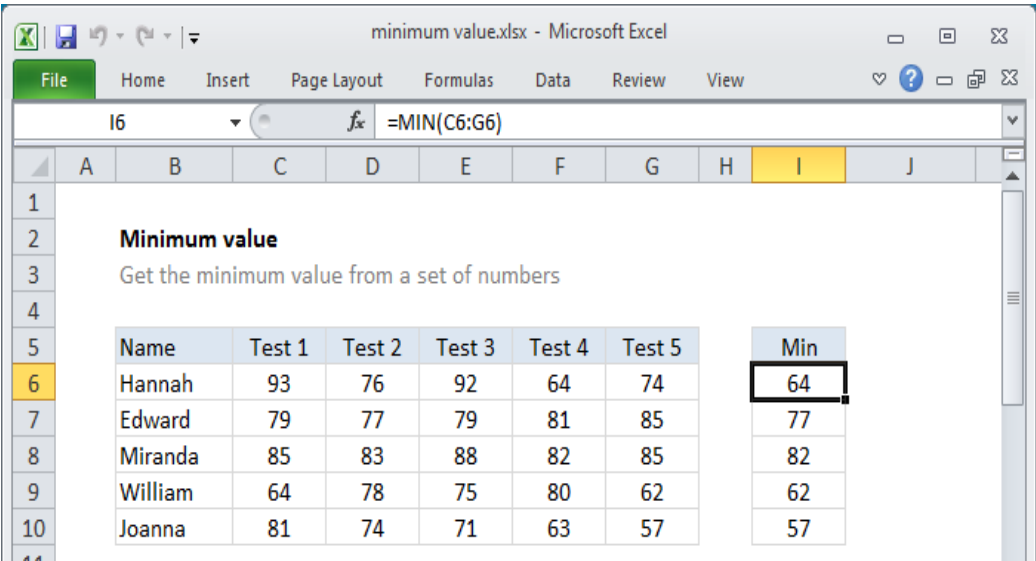
The Excel AVERAGE function calculates the average (arithmetic mean) of supplied numbers.

Name	Q1	Q2	Q3	Q4	Result
Jordi	84	67	55	89	74
Jean Luc	32	97	57	80	67
Beverly	95	87	63	94	85
Diana	92	52	94	81	80
Wesley	86	34	47	74	60
William	65	56	78	35	59
Miles	35	31	65	43	44

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Select a cell below or to the right of the numbers for which you want to find the smallest number. , click **Min (calculates the smallest)** or

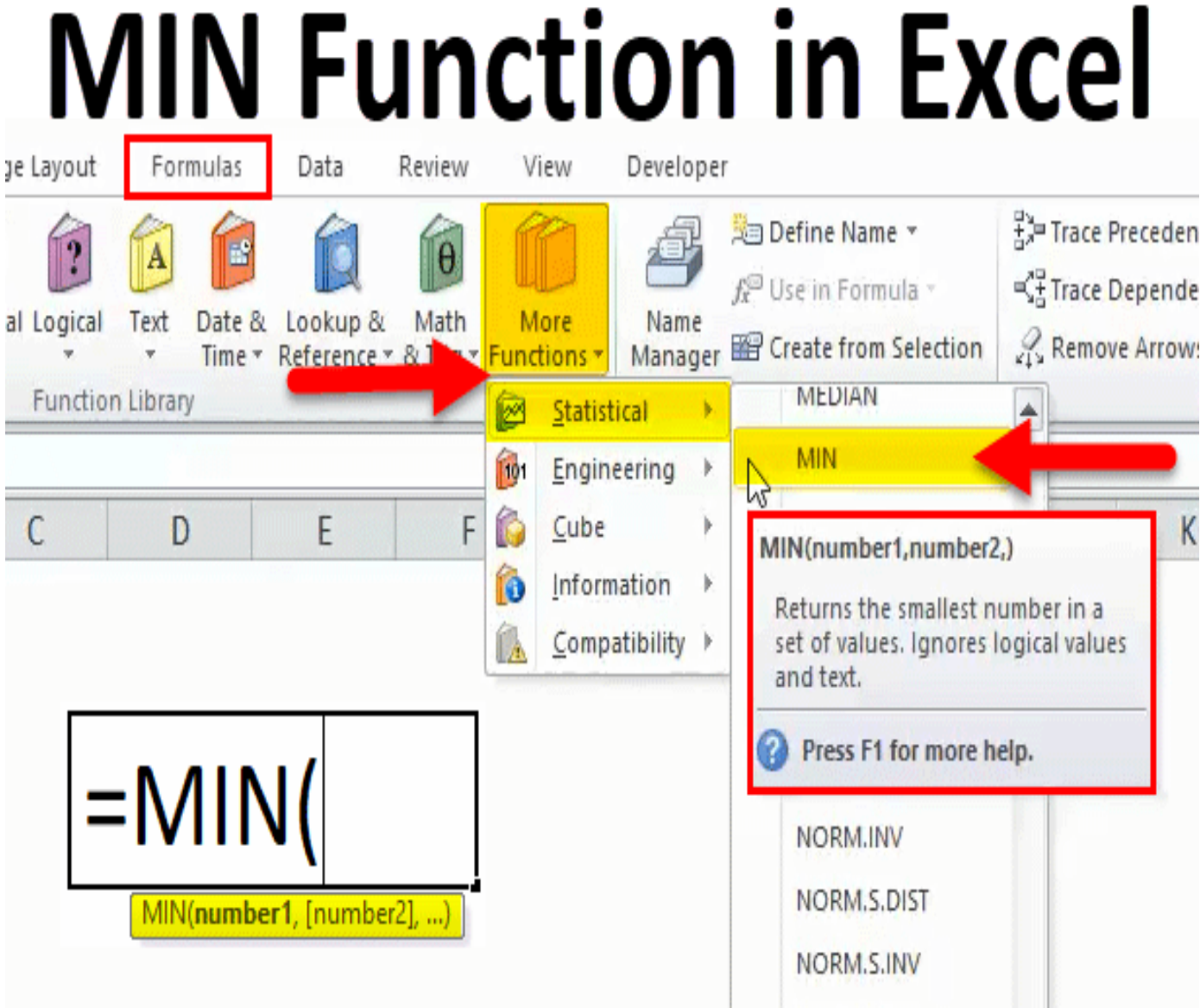
applying the MIN function formula =MIN(C6:G6) may give us the minimum value among the cell range from C6 to G6. It counts numbers but ignores empty cells, text, the logical values “TRUE” and “FALSE,” and text values.



The screenshot shows an Excel spreadsheet with the formula bar containing `=MIN(C6:G6)`. Below the formula bar, a table displays test scores for five students across five tests, with the minimum value calculated for each student.

Name	Test 1	Test 2	Test 3	Test 4	Test 5	Min
Hannah	93	76	92	64	74	64
Edward	79	77	79	81	85	77
Miranda	85	83	88	82	85	82
William	64	78	75	80	62	62
Joanna	81	74	71	63	57	57

MIN Function in Excel



The screenshot shows the Excel ribbon with the **Formulas** tab selected. The **More Functions** button is highlighted, and the **Statistical** category is expanded, showing the **MIN** function. A red box highlights the **MIN** function, and a tooltip provides the syntax and description.

MIN(number1,number2,...)

Returns the smallest number in a set of values. Ignores logical values and text.

Press F1 for more help.

MIN(number1, [number2], ...)

MAX FUNCTION

- The MAX function syntax has the following arguments: Number1, number2, ..Number1 is required, subsequent numbers are optional. 1 to 255 numbers for which you want to find the maximum value.

maximum value.xlsx - Microsoft Excel

File Home Insert Page Layout Formulas Data Review View

I6 fx =MAX(C6:G6)

Name	Test 1	Test 2	Test 3	Test 4	Test 5	Max
Hannah	93	76	92	64	74	93
Edward	79	77	79	81	85	85
Miranda	85	83	88	82	85	88
William	64	78	75	80	62	80
Joanna	81	74	71	63	57	81

Maximum value
Get the maximum value from a set of numbers

Sheet1

Ready

MAX Function in Excel

Page Layout Formulas Data Review View

Function Library

More Functions

Statistical

MAX

MAX(number1,number2,...)

Returns the largest value in a set of values. Ignores logical values and text.

Press F1 for more help.

MAX(number1, [number2], ...)

- Use the COUNT function to get the number of entries in a number field that is in a range or array of numbers. For example, you can enter the following formula to count the numbers in the range A1:A8: =COUNT(A2:A8). In this example, if five of the cells in the range contain numbers, the result is 5.

C3		<i>fx</i>	=COUNT(A2:A8)	
	A	B	C	D
1	Date			
2	5/1/2016			
3	1/6/2017		5	
4	9/13/2017			
5	25-25-2018			
6	9/13/2017			
7	14-85-2017			
8	5/7/2018			
9				

COUNT Formula in Excel

The screenshot shows the Excel ribbon with the 'Formulas' tab selected. In the 'Statistical' group, the 'COUNT' function is highlighted. A tooltip for the COUNT function is displayed, showing its syntax and description.

Function Library

CONFIDENCE.NORM
CONFIDENCE.T
CORREL
COUNT
COVARIANCE.P

COUNT
COUNT(value1,value2,...)
Counts the number of cells in a range that contain numbers.
Tell me more

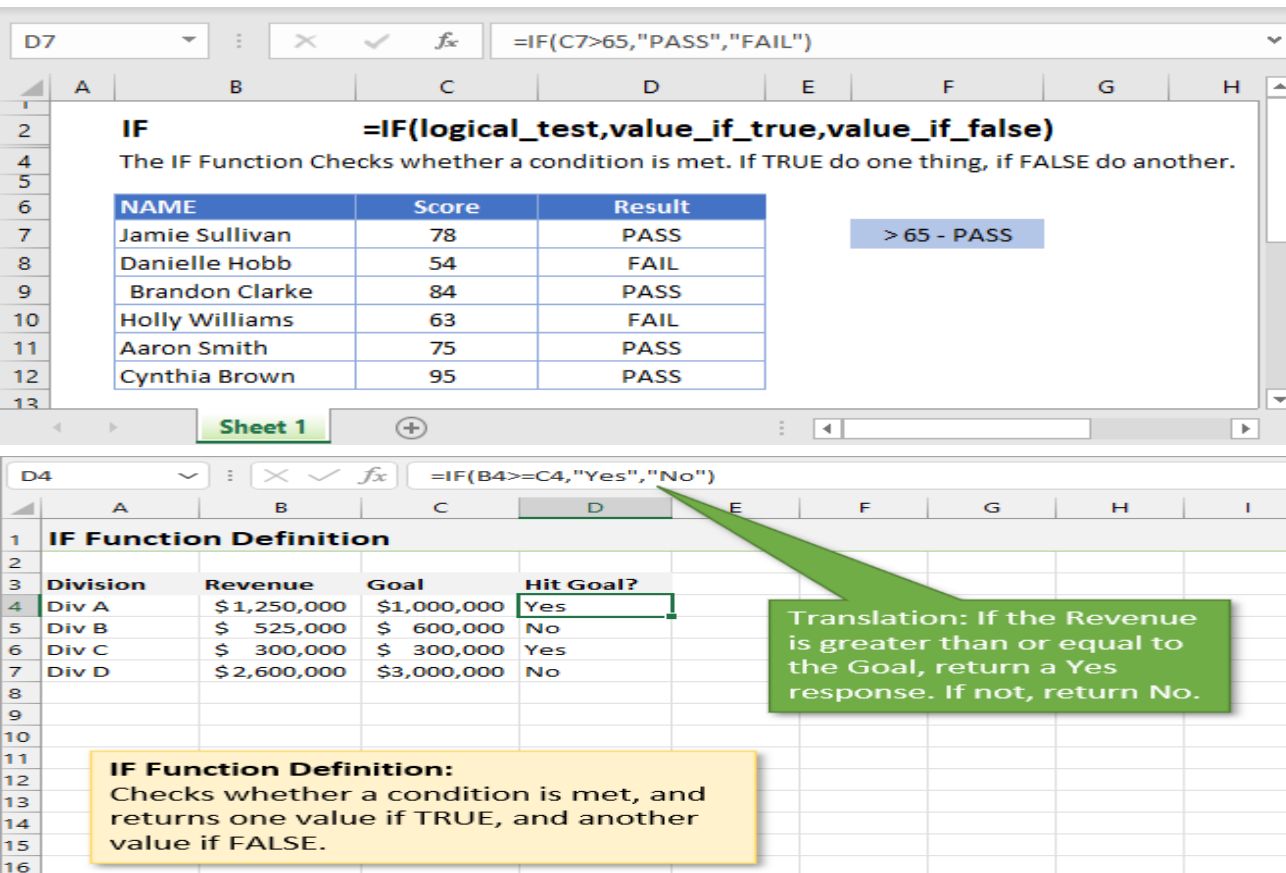
Worksheet Example:

	A	B	C	D	E	F
1	Values		Count			
2	CMI74		2			
3	LII					
4	EIJ					
5	5					
6	AUI					
7	-					

Formula Bar: =COUNT(

Formula Entry: COUNT(value1, [value2], ...)

- The IF function is one of the most popular functions in Excel, and it allows you to make logical comparisons between a value and what you expect. So an IF statement can have two results. The first result is if your comparison is True, the second if your comparison is False.



IF
The IF Function Checks whether a condition is met. If TRUE do one thing, if FALSE do another.

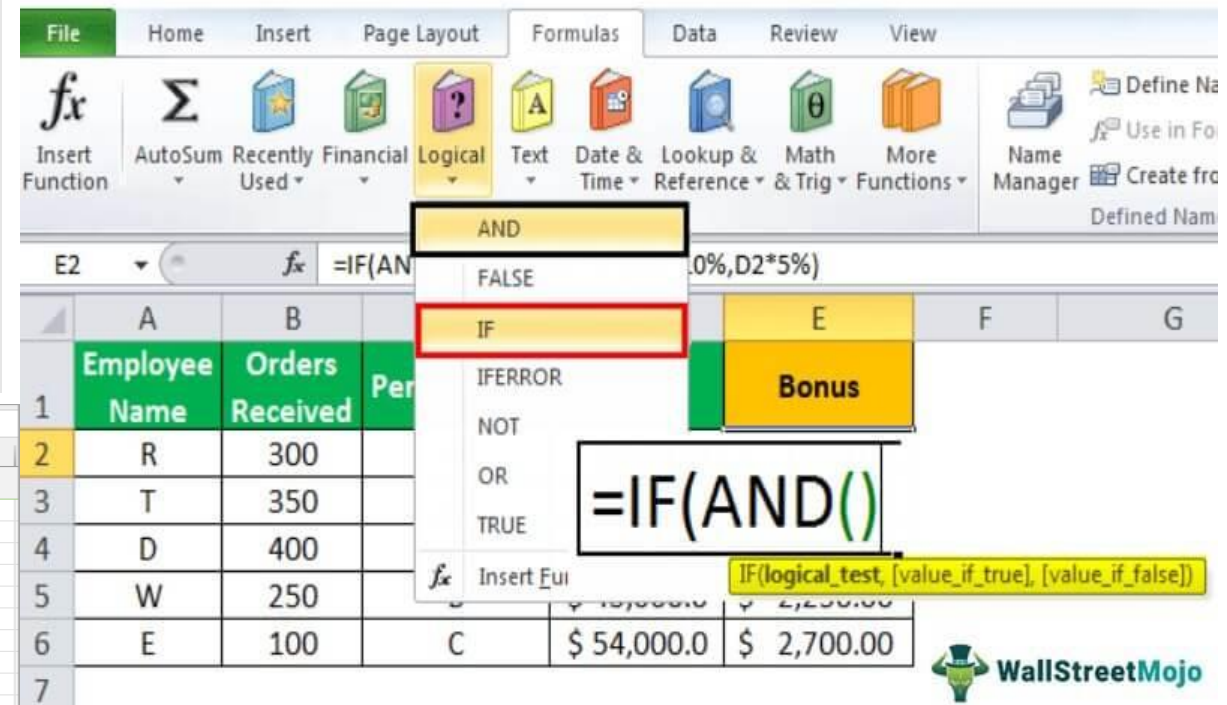
Formula: `=IF(logical_test,value_if_true,value_if_false)`

NAME	Score	Result
Jamie Sullivan	78	PASS
Danielle Hobb	54	FAIL
Brandon Clarke	84	PASS
Holly Williams	63	FAIL
Aaron Smith	75	PASS
Cynthia Brown	95	PASS

IF Function Definition:
Checks whether a condition is met, and returns one value if TRUE, and another value if FALSE.

Translation: If the Revenue is greater than or equal to the Goal, return a Yes response. If not, return No.

IF AND Formula in Excel



Formula: `=IF(AND(logical_test1,logical_test2,...),[value_if_true],[value_if_false])`

Employee Name	Orders Received	Per	Bonus
R	300		
T	350		
D	400		
W	250		
E	100	C	\$ 2,700.00

- Nested IF functions, meaning one IF function inside of another, allows you to test multiple criteria and increases the number of possible outcomes. We want to determine a student's grade based on their score. If Bob's score in B2 is greater than or equal to 90, return an A.

The IF Function has 3 arguments:

- Logical test. This is where we can compare data or see if a condition is met.
- Value if true. Defining this argument tells Excel to return a certain value if the condition in the logical test is met.
- Value if false.

Nested IF Condition in Excel

C2 : X ✓ fx **=IF(B2>=585, "Dist",IF(B2>=500, "First",IF(B2>=400, "Second",IF(B2>=300, "Pass", "Fail"))))**

	A	B	C	D	E	F
1	Student	Score	Result			
2	A	271	Fail			
3	B	320	Pass			
4	C	271	Fail			
5	D	229	Fail			
6	E	590				
7	F	456				
8	G	468				
9	H	483				
10	I	514	First			
11	J	502	First			
12						
13						
14						

=IF(

IF(logical_test, [value_if_true], [value_if_false])

Nested IF Formula in Excel

Excel ribbon showing the Formula tab. The formula bar displays the following nested IF formula:

```
=IF(C3>=85,"DISTINCTION",IF(C3>=60,"FIRST CLASS",IF(C3>=35,"SECOND CLASS","FAIL")))
```

	A	B	C	D	E	F	G	H
1								
2								
3								
4								
5								
6								
7								
8								

Name	Final_Score	Result
Foakes	31	FAIL
Burns	48	SECOND CLASS
Jimmy	82	FIRST CLASS
Tom	33	FAIL
Sam	38	SECOND CLASS
Curran	49	SECOND CLASS

Name	Final_Score	Result
Foakes	31	=IF(C3>=85, "DISTINCTION", IF(C3>=60, "FIRST CLASS", "SECOND CLASS", "FAIL"))
Burns	48	
Jimmy	82	
Tom	33	
Sam	38	

Excel formula bar showing the following nested IF formula:

```
=IF(D2="USA",E2*10%,IF(D2="Australia",E2*5%,IF(D2="Singapore",E2*2%)))
```

	A	B	C	D	E	F
1	Month Sold	Brand	Model	Sales Location	Sale Price	Commission Fee
2	JAN	Samsung	Galaxy S9	Australia	\$1,000	50
3	JAN	Samsung	Galaxy S8	USA	\$800	80
4	JAN	Apple	IPhone7	Singapore	\$650	13
5	JAN	Apple	IPhone8	Australia	\$850	42.5
6	JAN	Apple	IPhoneXS	USA	\$1,400	140
7	JAN	Samsung	Galaxy S9	Australia	\$1,000	50
8	JAN	Samsung	Galaxy S9	USA	\$1,000	100

Excel formula bar showing the following nested IF formula:

```
=IF(AND(B5="red",C5>=100),1.5,IF(AND(B5="red",C5<100),1.4,IF(AND(B5="blue",C5>=100),1.3,IF(AND(B5="blue",C5<100),1.2,1.1))))
```

Nested IF with multiple AND

Color	Value	Result
Red	101	1.5
Red	99	1.4
Blue	103	1.3
Blue	96	1.2
Green	101	1.1

Excel formula bar showing the following nested IF formula:

```
=IF(B2>249,"Excellent",IF(B2>=200,"Good",IF(B2>150,"Satisfactory","Poor")))
```

	A	B	C	D	E	F
1	Student name	Score	Result			Scores
2	Brian	274	Excellent		Excellent	Over 249
3	Christian	280	Excellent		Good	249 to 200
4	Colin	170	Satisfactory		Satisfactory	199 to 150
5	Laura	237	Good		Poor	Under 150
6	Melissa	210	Good			
7	Mike	186	Satisfactory			
8	Neal	240	Good			
9	Peter	146	Poor			
10	Rachel	287	Excellent			

ALGAE

For example, DATE(2008,1,2) returns January 2, 2008. If year is less than 0 or is 10000 or greater, Excel returns the #NUM!

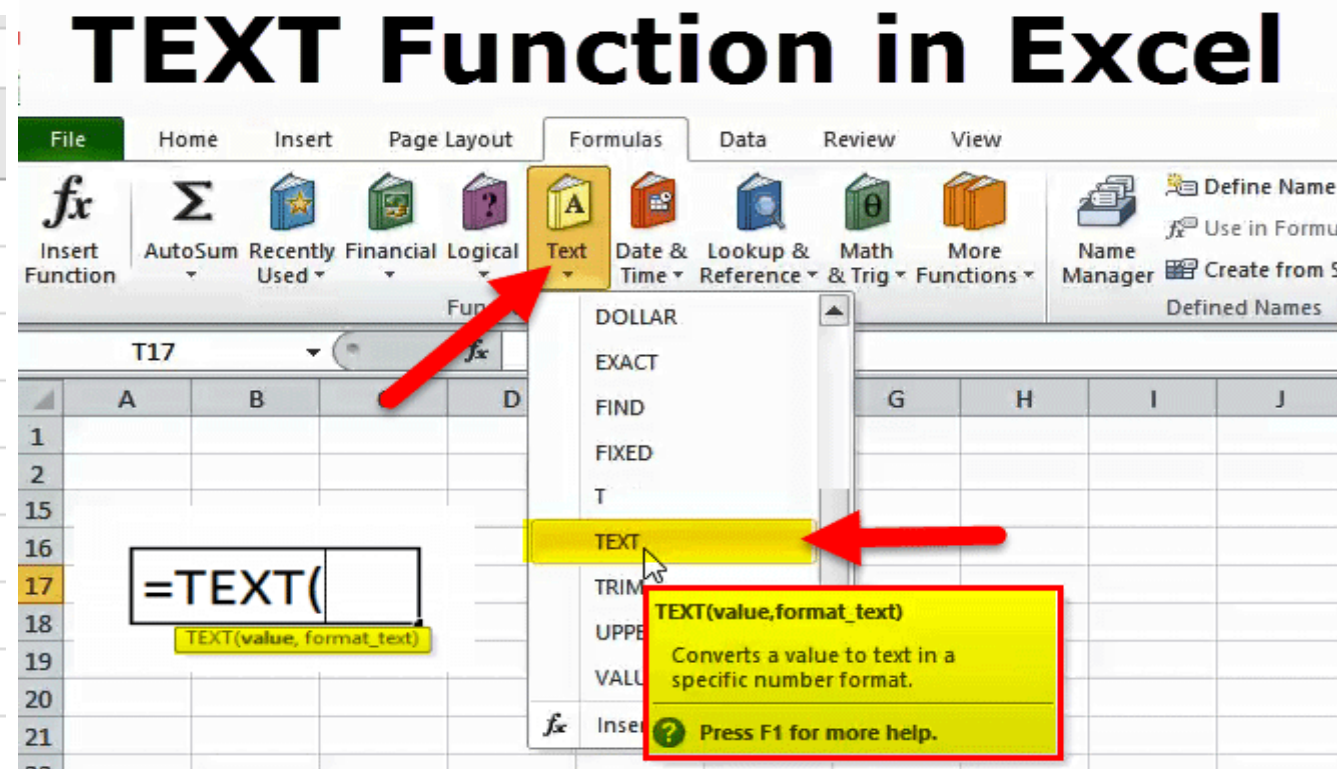
E5					=DATE(B5,C5,D5)
	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

The screenshot shows an Excel spreadsheet with columns A, B, C, and D, and rows 1 through 10. In cell B4, the formula `=DATE(` is entered. A tooltip is displayed over the formula bar, showing the function name **DATE(year,month,day)** and its description: "Returns the number that represents the date in Microsoft Excel date-time code." Below the description, there is a link that says "Press F1 for more help." The tooltip is highlighted with a red border.

- The TEXT function lets you change the way a number appears by applying formatting to it with format codes. It's useful in situations where you want to display numbers in a more readable format, or you want to combine numbers with text or symbols.
- Format_text: A text string that defines the format...
- Value: A numeric value that you want to be con...

C4				=TEXT(B4,"dd-mm-yy")
	A	B	C	D
1				
2				
3		Nummber	Formated	Formula
4		7/16/2019	16-07-19	=TEXT(A2,"dd-mm-yy")
5		10000	\$ 10000	=TEXT(A3,"\$ 0")
6		2124	02124	=TEXT(A4,"00000")
7		23	00023	=TEXT(A5,"00000")
8				

TEXT Function in Excel



The screenshot shows the Excel ribbon with the Formulas tab selected. The Text function is highlighted in the dropdown menu. A red arrow points to the Text function, and another red arrow points to the TEXT function in the dropdown list. A yellow box highlights the TEXT function description: "TEXT(value,format_text) Converts a value to text in a specific number format. Press F1 for more help."

- The UPPER Function[1] is an Excel Text function, that will convert text to all capital letters (UPPERCASE). Thus, the function converts all characters in a supplied text string into upper case. In financial analysis, we often import data from external sources.

C3			=UPPER(B3)
	A	B	C
1			
2		Text	Result
3		ExCel fUNctIon	EXCEL FUNCTION
4		Smith1234	SMITH1234
5		Smith123*	SMITH123*
6		to be or not to be	TO BE OR NOT TO BE
7		ALPHABET	ALPHABET

UPPER Function Overview			
	Text	With Upper Cases	
4	washington	WASHINTON	=UPPER("washinton")
5	Washington dc	WASHINGTON DC	=UPPER("Washington dc")
6	64, park street	64, PARK STREET	=UPPER("64, park street")
7	andrew mcdonald	ANDREW MCDONALD	=UPPER("andrew mcdonald")
8	\$5,000	5000	=UPPER(B9) // Affected
9	8/19/2021	44427	=UPPER(B10) // Affected
10	433.455	433.455	=UPPER(B11) // Not Affected

Uppercase in Excel

FileHomeInsertPage LayoutFormulasDataReviewViewDeveloper

fxInsert Function

ΣAutoSum

Recently Used

Financial

Logical

Text

Date & Time

Lookup & Reference

Math & Trig

More Functions

Name Manager

Define New

Use in Formula

Create from Defined Name

fx=UPPER(A2)

	A	B	C
1	Name	Age	Uppercase
2	jOhn	23	JOHN
3	Smith	25	SMITH
4	Jackson	32	JACKSON
5	fisCher	24	FISCHER
6	ColBy p Ford	29	COLBY P FORD
7	emma	30	EMMA
8	aleXANder	22	ALEXANDER
9	Suzzane	26	SUZZANE

DOLLAR
EXACT
FIND
FIXED
LEFT
LEN
UPPER
VALUE
Insert Function...

UPPER(text)
Converts a text string to all uppercase letters.
Press F1 for more help.

- EXCELJET** 

Lower Case in Excel

The screenshot displays two Excel workbooks. The top workbook, titled 'PRODUCT NAMES', shows a table with 'Verbs' in column A and their lowercase equivalents in column B. A formula bar shows '=LOWER(A2)'. A callout bubble labeled 'Excel Function' points to the formula bar. A red box highlights the 'Convert To Lower Case' button in the ribbon. A callout bubble labeled 'Command Button' points to the 'Convert To Lower Case' button. The bottom workbook, titled 'Groups in Formula Tab', shows a table with 'Groups in Formula Tab' in column A and their lowercase equivalents in column B. A callout bubble labeled 'Flash Fill' points to the Flash Fill icon in the ribbon. A red box highlights the 'Accept suggestions' button in the ribbon.

Verbs	Lower Case
EAT	eat
WRITE	write
SING	sing

Groups in Formula Tab	Lower Case
FUNCTION LIBRARY	function library
DEFINED NAMES	defined names
FORMULA AUDITING	formula auditing
CALCULATION	calculation

D5 =SQRT(B5)

SQRT Function		
NUMBER	FORMULA USED	RESULT
25	=SQRT(B5)	5
49	=SQRT(B6)	7
900	=SQRT(B7)	30
121	=SQRT(B8)	11
100	=SQRT(B9)	10

C5 =SQRT(B5)

SQRT function	
Number	Result
529	23
289	17
225	15
121	11
89	9.433981
49	7
9	3
1/4	1/2
1/9	1/3
0	0
-4	#NUM!

B2 =SQRT(A2)

	A	B
1	Number	Sqaure root
2	25	5
3	16	4
4	9	3
5	4	2
6	0	0
7	-4	#NUM!
8	-9	#NUM!

Square Root Function in Excel

File Home Insert Page Layout **Formulas** Data Review View Developer

Function Library

J19

	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					

=SQRT(

SQRT(number)

ROUNDUP

SERIESSUM

SIGN

SIN

SINH

SQRT

SQRTPI

SUBTOTAL

SUM

SUMIF

SUMIFS

SQRT(number)

Returns the square root of a number.

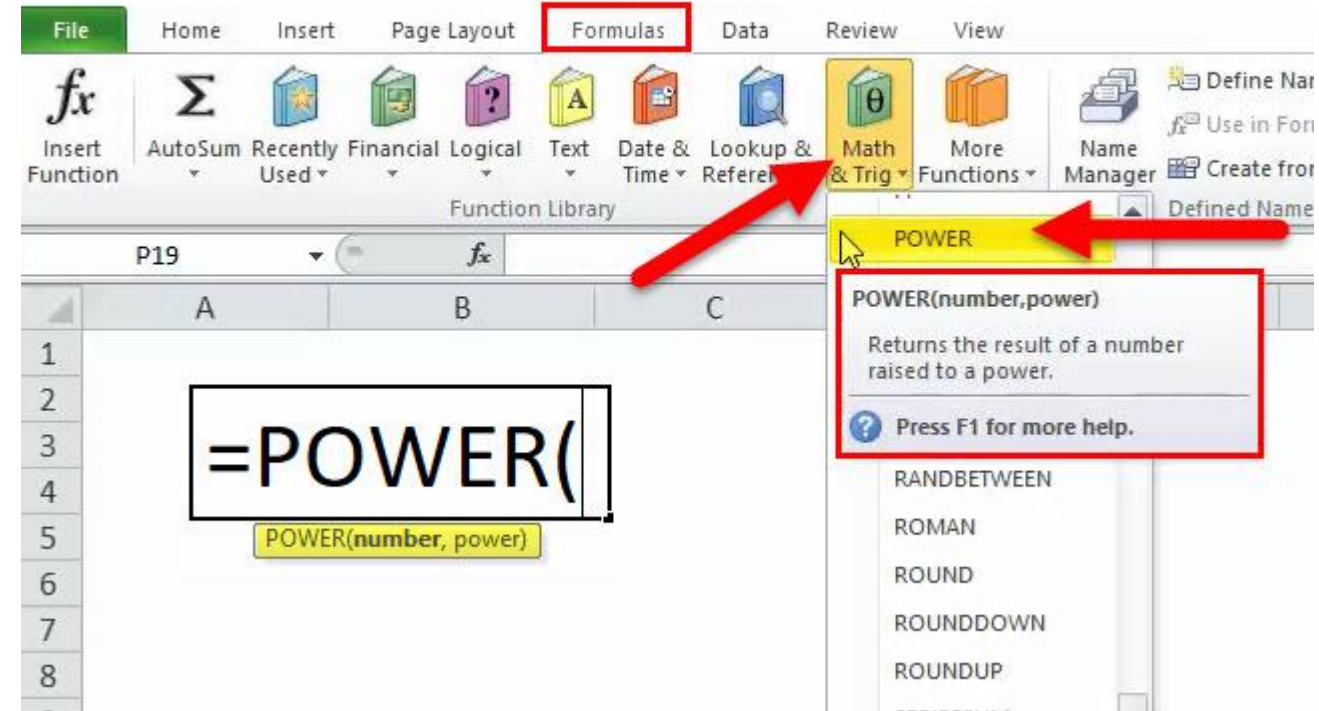
Press F1 for more help.

The Excel POWER function returns a number raised to a given power. The POWER function is an alternative to the exponent operator (^).

- Raise a number to a power.
- Number raised to power.
- =POWER (number, power)
- number - Number to raise to a power.
- power - Power to raise number to (the exponent).

POWER(number, power)		
nth power of		
number	power	POWER
5	2	25
4	3	64
0	4	0
-3	5	-243
-7	6	117649

POWER Function in Excel



- [Arithmetic Operators](#)
- [Logical/Comparison Operators](#)
- [Text Concatenation Operator](#)
- [Reference Operators](#)

• <https://www.brainbell.com/tutorials/Excel-2007/operators-in-formulas.html>

Operators in Excel

Operator	Condition Name	Excel Formula	Description
+	Addition	= A1 + B1	Makes addition of two or more than two numeric values and gives a numeric value as a return
-	Subtraction	= A1 - B1	Subtracts two or more than two numeric values and gives a numeric value in return
<>	Not Equals To	=A1 <> B1	Multiplies one numeric value with the other and gives a numeric value in return
=	Equals To	=A1 = B1	Divides the numerator(should be numeric) value with the denominator(should be numeric) value and gives a numeric value in return
:	Range	= SUM(A2:B6)	Converts the number present in a cell into a percentage (i.e. divides the number by 100)
,	Union	= SUM(A2:A6, A2:B3)	Raises the power of first numeric value by second numeric value

- If a formula contains operators with the same precedence — for example, if a formula contains both a multiplication and division operator — Excel evaluates the operators from left to right.

	A	B	C
1	Symbol	Operator	Precedence
2	:	Range	1
3	<Space>	Intersection	2
4	,	Union	3
5	-	Negation	4
6	%	Percentage	5
7	^	Exponential Operator	6
8	*	Multiplication	7
9	/	Division	7
10	+	Addition	8
11	-	Subtraction	8
12	&	Concatenation	9
13	=	Equal to	10
14	>	Greater than	10
15	<	Less than	10
16	>=	Greater than Equal to	10
17	<=	Less than Equal to	10
18	<>	Not Equal to	10

B5

✕ ✓ f_x

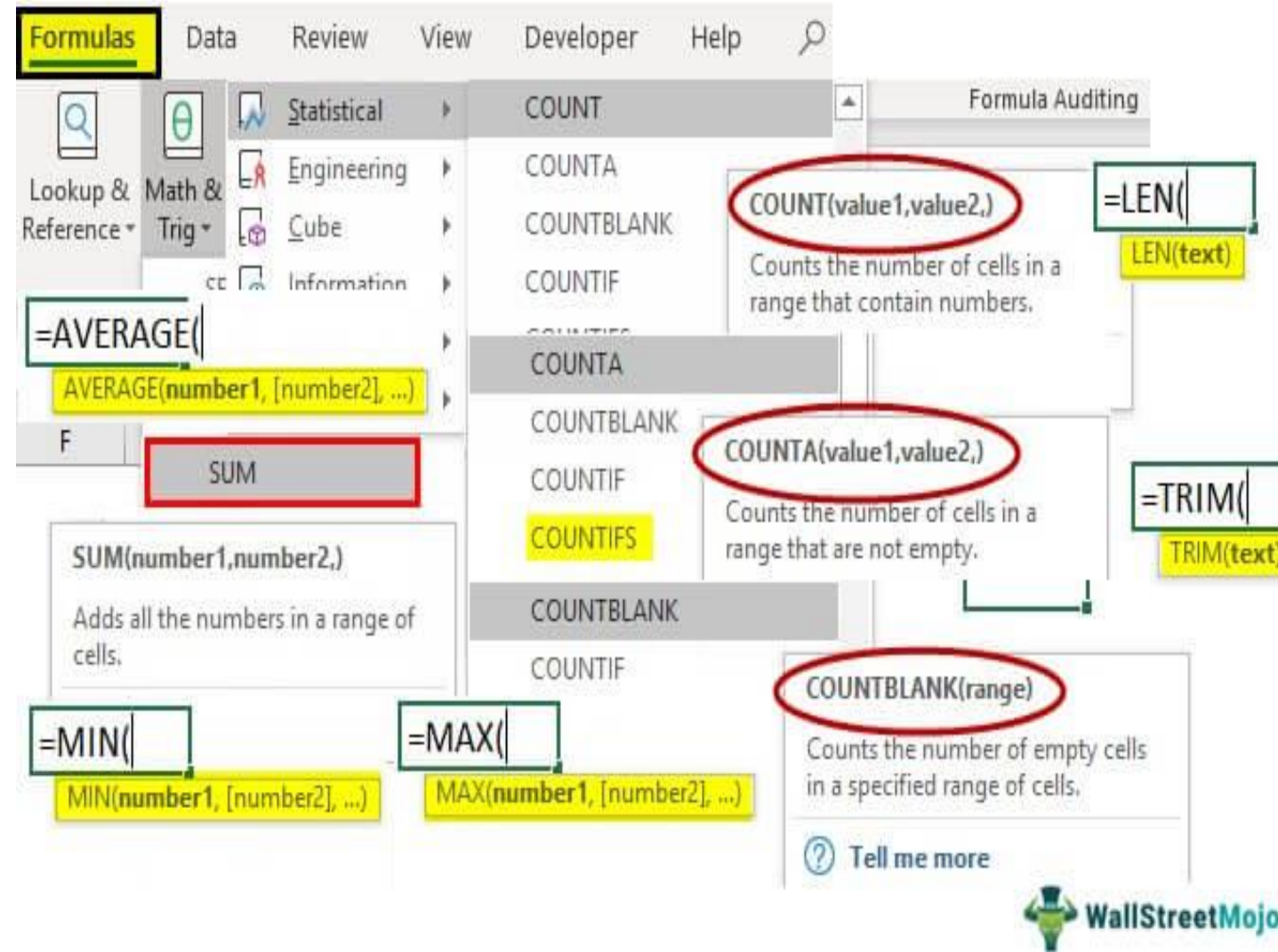
1

	A	B	C	D	E	F
1	Order of Operations					
2						
3						
4						
5		Order	Symbols	Operation	Example	
6		1	()	Parentheses	=(5-2)*4 = 12	
7		2	:,	Reference operators	=SUM(A1:A5)	
8		3	-	Negation	=-3^2 = 9	
9		4	%	Percent	=5%*100 = 5	
10		5	^	Exponentiation	=5^2*2 = 50	
11		6	*/	Multiplication and Division	=7-6/2 = 4	
12		7	+ -	Addition and Subtraction	=6/2+1 = 4	
13		8	&	Concatenation	="score: "&5+1 = score: 6	
14		9	> < = <>	Logical comparisons	=3^2>5+3 = TRUE	
15						
16						

EXCELJET

- Enter a formula that contains a built-in function
- Select an empty cell.
- Type an equal sign = and then type a function. For example, =SUM for getting the total sales.
- Type an opening parenthesis (.
- Select the range of cells, and then type a closing parenthesis).
- Press Enter to get the result.

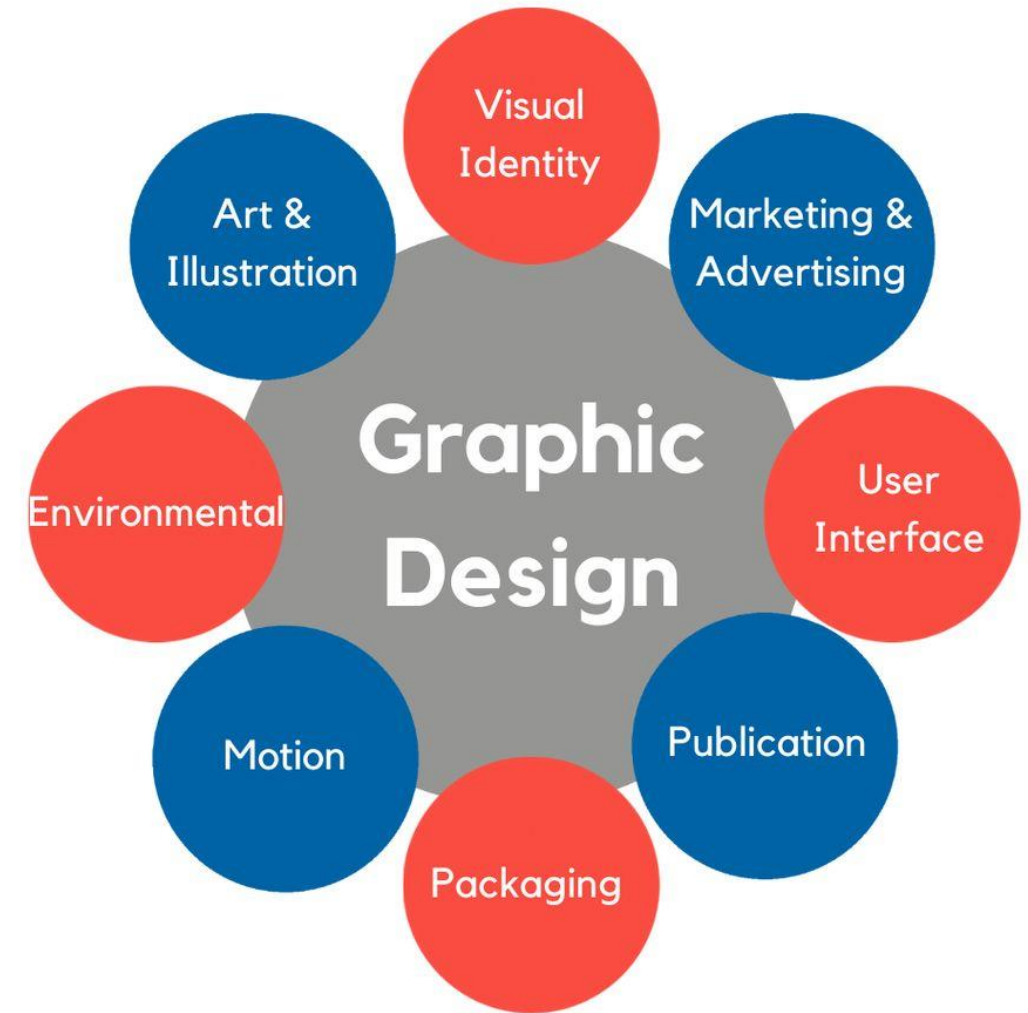
Basic Excel Formulas



The screenshot displays the Microsoft Excel ribbon, specifically the **Formulas** tab. The ribbon includes sub-tabs for **Statistical**, **Engineering**, **Cube**, and **Information**. The **Statistical** sub-tab is active, showing a list of functions: **COUNT**, **COUNTA**, **COUNTBLANK**, **COUNTIF**, **COUNTIFS**, **COUNTBLANK**, and **COUNTIF**. The **COUNT** function is highlighted, and its description is shown: "COUNT(value1,value2.) Counts the number of cells in a range that contain numbers." Other functions like **COUNTA**, **COUNTBLANK**, **COUNTIF**, **COUNTIFS**, and **COUNTBLANK** are also listed with their descriptions. The **SUM** function is highlighted in the ribbon, and its description is shown: "SUM(number1,number2,) Adds all the numbers in a range of cells." The **MIN** and **MAX** functions are also highlighted with their descriptions: "MIN(number1, [number2], ...) Minimum value" and "MAX(number1, [number2], ...) Maximum value". The **LEN** function is highlighted with its description: "LEN(text) Length of text". The **TRIM** function is highlighted with its description: "TRIM(text) Trims text". The WallStreetMojo logo is visible in the bottom right corner.

Types of graphic design

- Product design. Product designers use their creative prowess to research, design and develop new products. ...
 - Branding design. ...
 - Website design. ...
 - Print design. ...
 - Publishing design. ...
 - Environmental design. ...
 - Animation design.
- Graphics** are visual images or designs on some surface, such as a wall, canvas, screen, paper, or stone, to inform, illustrate, or entertain.



Filling Word Art or Auto Shapes with Graphics

Create the AutoShape you want.

- On the Drawing toolbar, click the arrow to the right of the **Fill Color** button 
- Click the **Picture** tab
- Click the **Select Picture** button.
- Browse to where the picture is saved; select the picture; click the **Insert** button
- Click **OK**

To make the picture transparent (as in Figure 4)

- Double click the filled object
- Select the **Colors and Lines** tab **Figure 5: Picture**
- Slide the **Transparency** button to the right

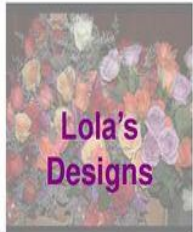


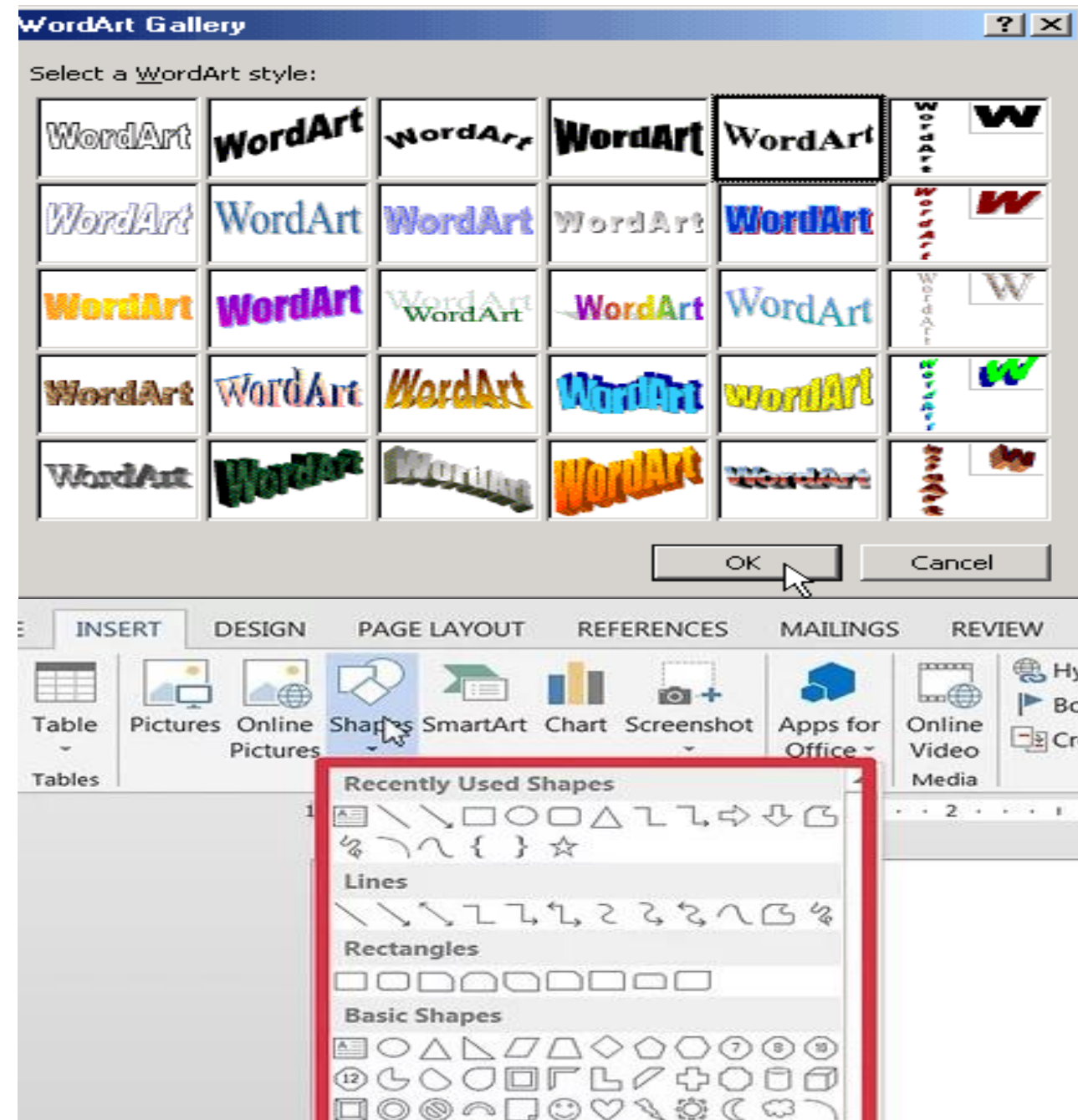
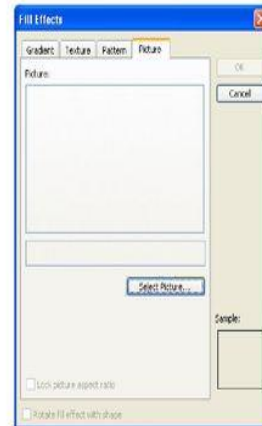
Figure 4: Text Box (picture fill, transparency 50%, lines removed)



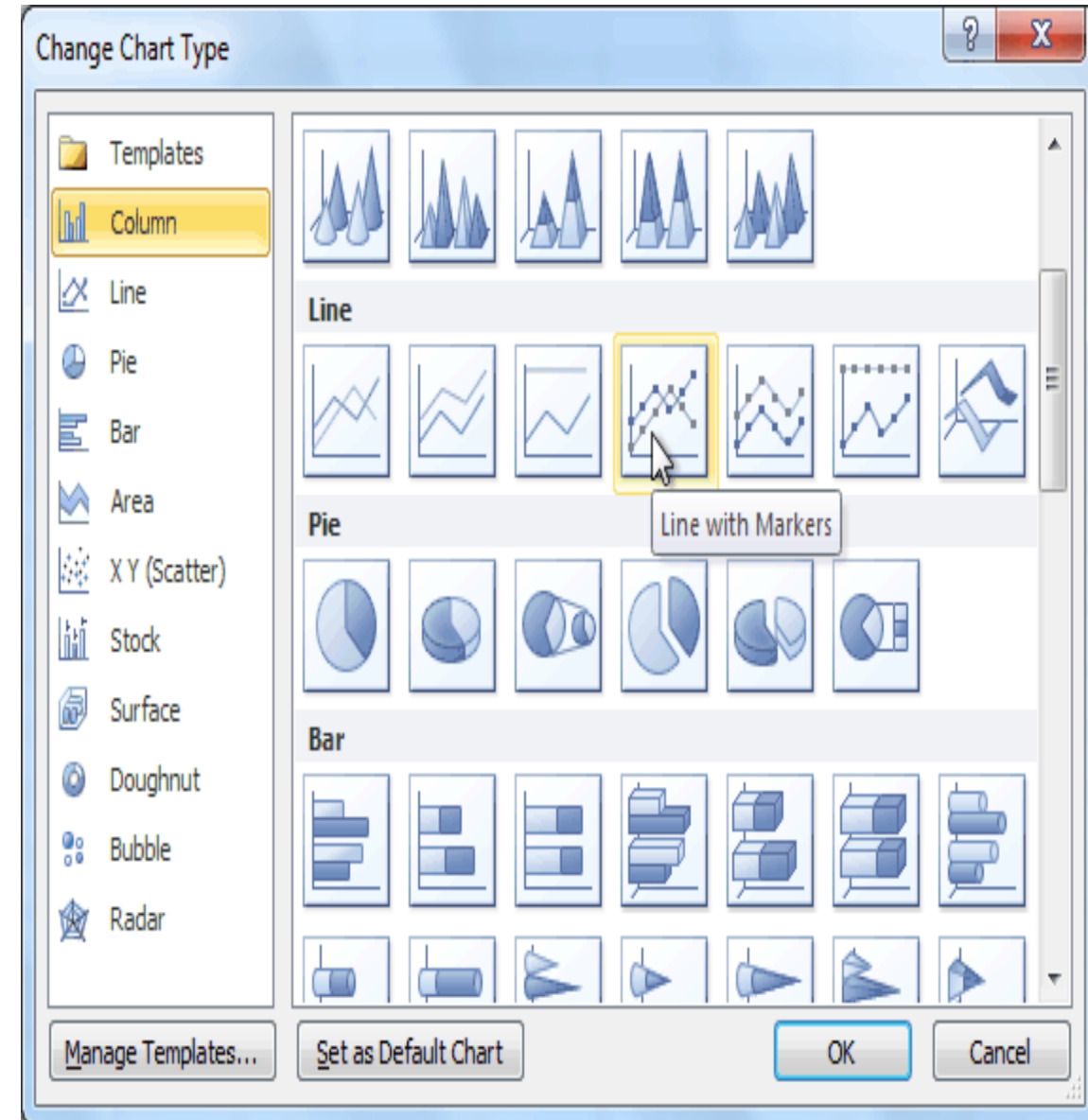
Figure 5: Picture



Figure 6: Word Art with Picture Fill



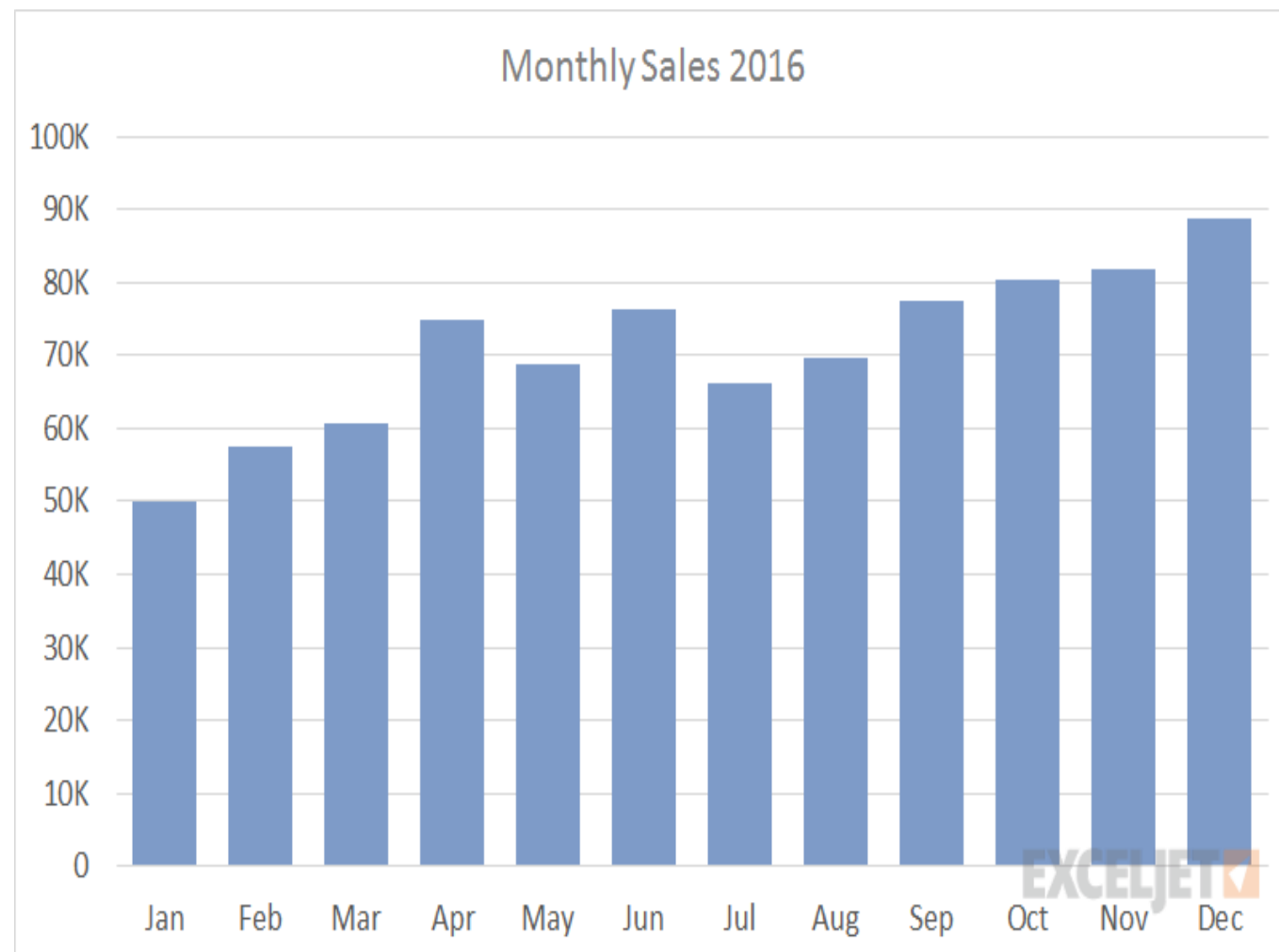
- In Microsoft Excel, charts are used to make a graphical representation of any set of data. A chart is a visual representation of data, in which the data is represented by symbols such as bars in a bar chart or lines in a line chart.
- Column Chart
- Line Chart
- Pie Chart
- Bar Chart
- Area Chart
- XY (Scatter) Chart
- Bubble Chart
- Stock Chart
- Surface Chart
- Doughnut
- Radar Chart
- Combo Chart



A Column Chart typically displays the categories along the horizontal (category) axis and values along the vertical (value) axis. To create a column chart, arrange the data in columns or rows on the worksheet.

A column chart has the following sub-types

- Clustered Column.
- Stacked Column.
- 100% Stacked Column.
- 3-D Clustered Column.
- 3-D Stacked Column.
- 3-D 100% Stacked Column.
- 3-D Column.

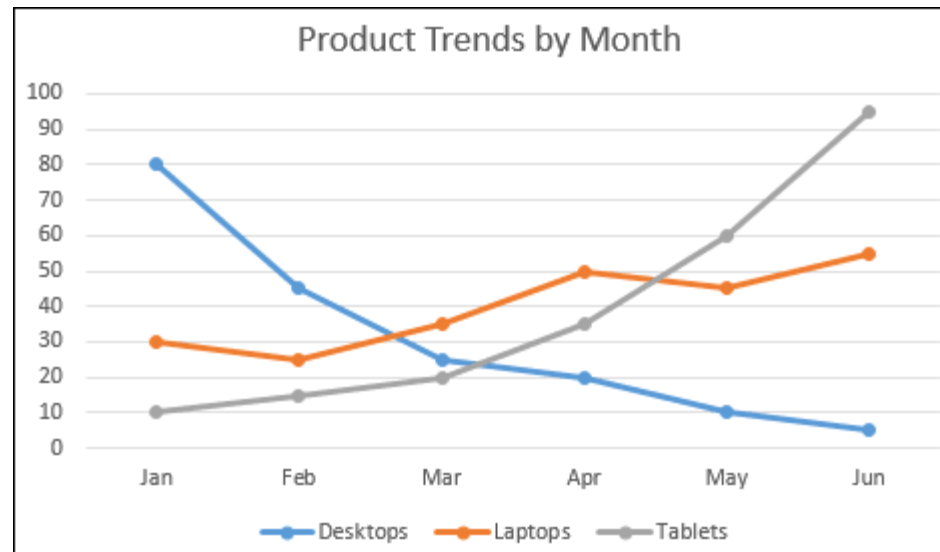


- Line charts can show continuous data over time on an evenly scaled Axis. Therefore, they are ideal for showing trends in data at equal intervals, such as months, quarters or year.

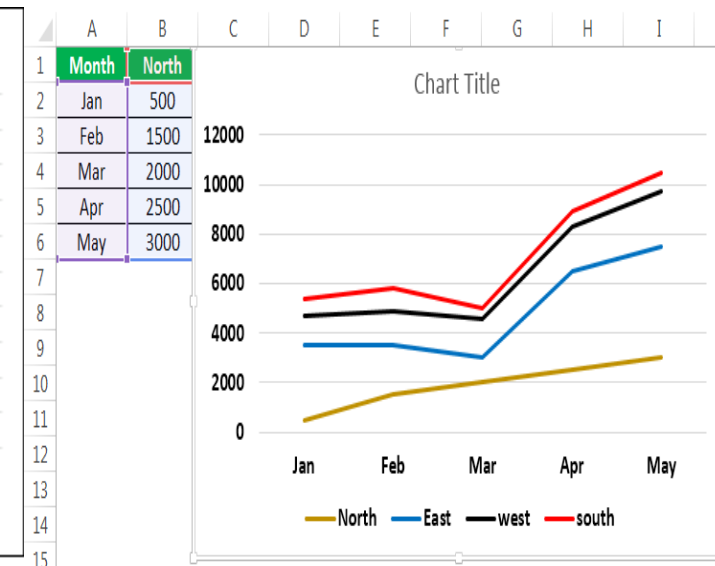
In a Line chart –

- Category data is distributed evenly along the horizontal axis.
- Value data is distributed evenly along the vertical axis.
- To create a Line chart, arrange the data in columns or rows on the worksheet.
- A Line chart has the following sub-types

- Stacked Line
- 100% Stacked Line
- Line with Markers
- Stacked Line with Markers
- 100% Stacked Line with Markers
- 3-D Line



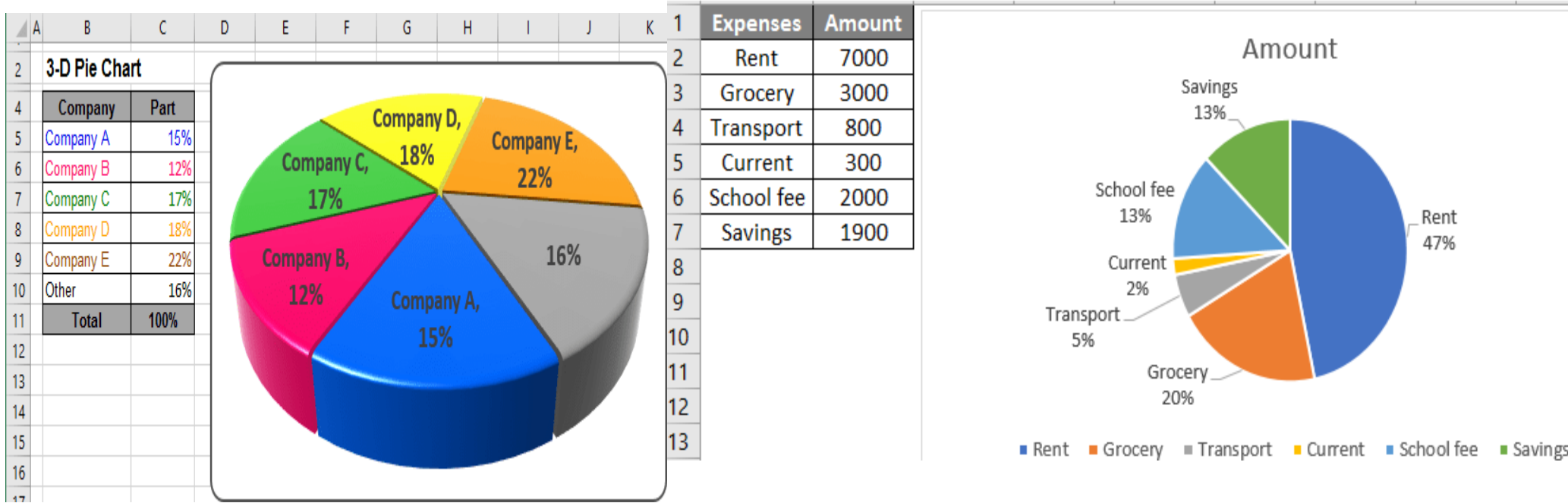
TechOnTheNet.com



Pie charts show the size of items in one data series, proportional to the sum of the items. The data points in a pie chart are shown as a percentage of the whole pie. To create a Pie Chart, arrange the data in one column or row on the worksheet.

- A Pie Chart has the following sub-types –
- Pie
- 3-D Pie
- Pie of Pie
- Bar of Pie

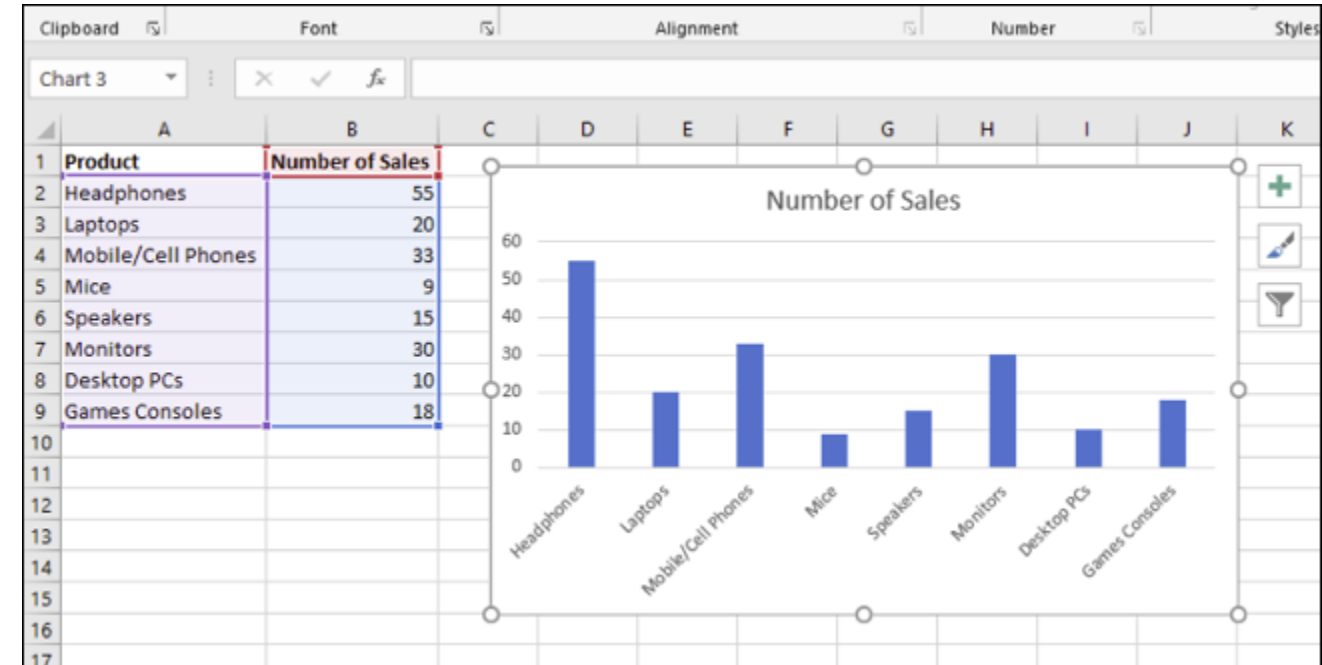
Pie Chart Examples



Bar Charts illustrate comparisons among individual items. In a Bar Chart, the categories are organized along the vertical axis and the values are organized along the horizontal axis. To create a Bar Chart, arrange the data in columns or rows on the Worksheet.

A Bar Chart has the following sub-types –

- Clustered Bar
- Stacked Bar
- 100% Stacked Bar
- 3-D Clustered Bar
- 3-D Stacked Bar
- 3-D 100% Stacked Bar

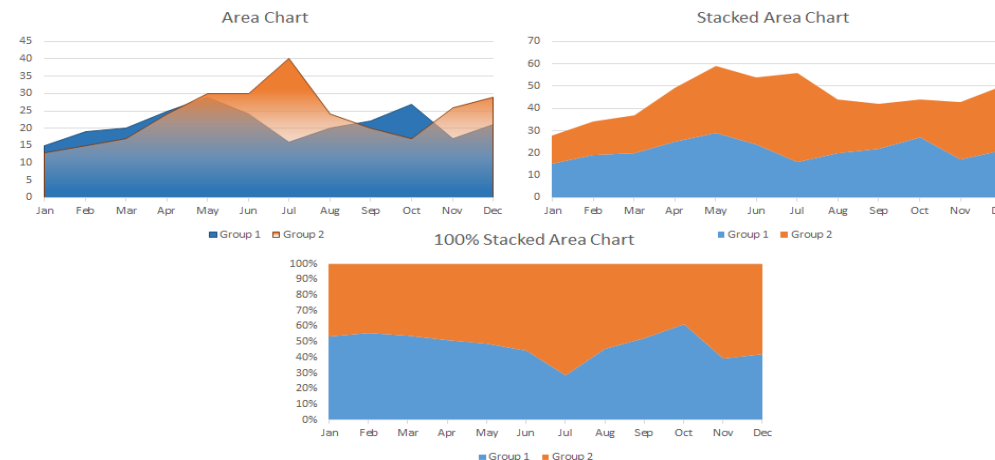
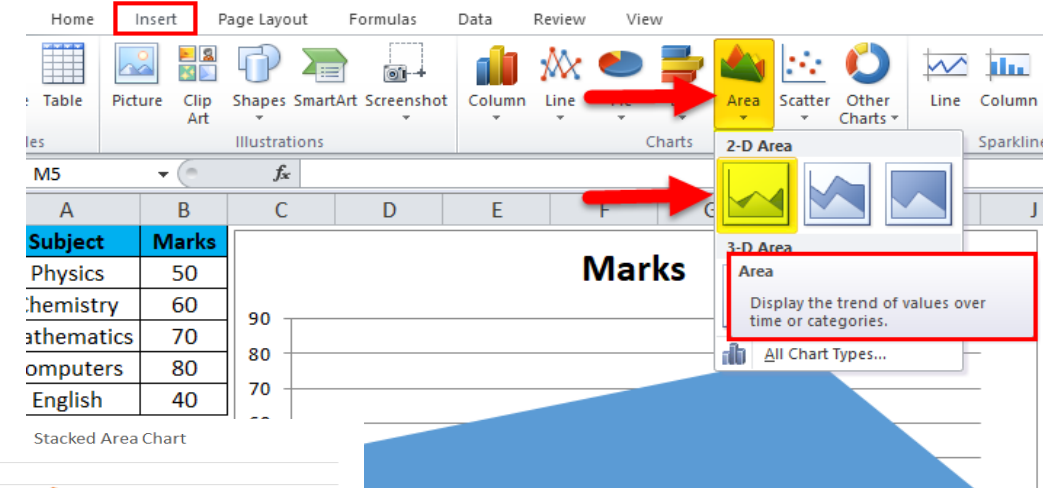


Area Charts can be used to plot the change over time and draw attention to the total value across a trend. By showing the sum of the plotted values, an area chart also shows the relationship of parts to a whole. To create an Area Chart, arrange the data in columns or rows on the worksheet.

An Area Chart has the following sub-types –

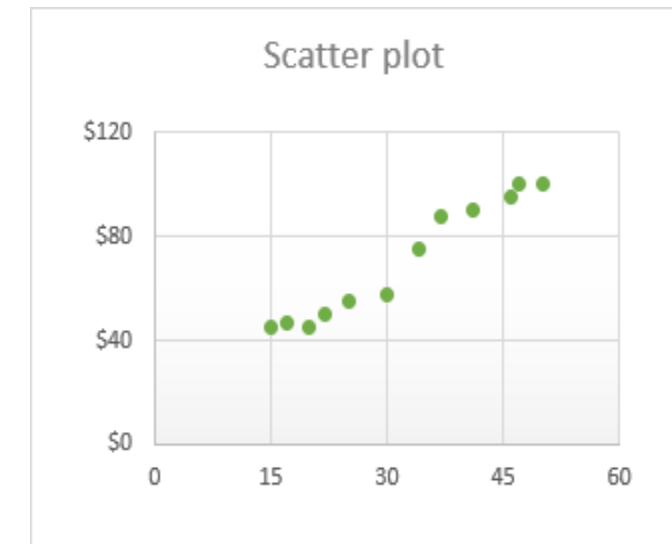
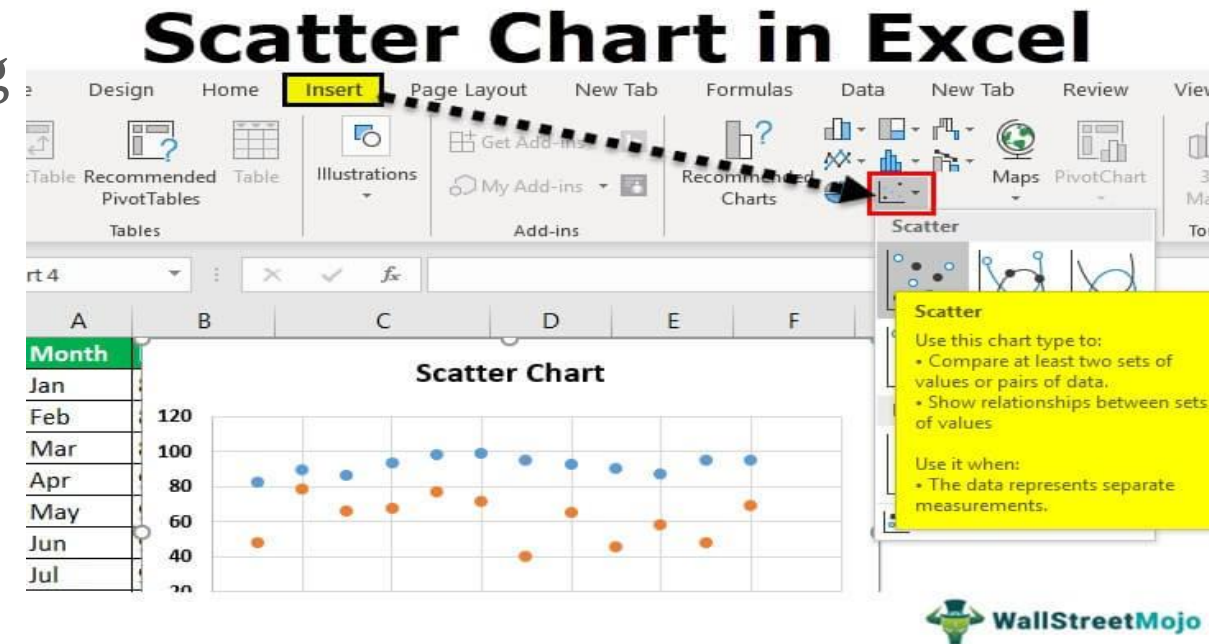
- Area
- Stacked Area
- 100% Stacked Area
- 3-D Area
- 3-D Stacked Area
- 3-D 100% Stacked Area

Area Chart in Excel



XY (Scatter) charts are typically used for showing and comparing numeric values, like scientific, statistical, and engineering data.

- A Scatter chart has two Value Axes –
- Horizontal (x) Value Axis
- Vertical (y) Value Axis
- It combines x and y values into single data points and displays them in irregular intervals, or clusters. To create a Scatter chart, arrange the data in columns and rows on the worksheet.
- Place the x values in one row or column, and then enter the corresponding y values in the adjacent rows or columns.

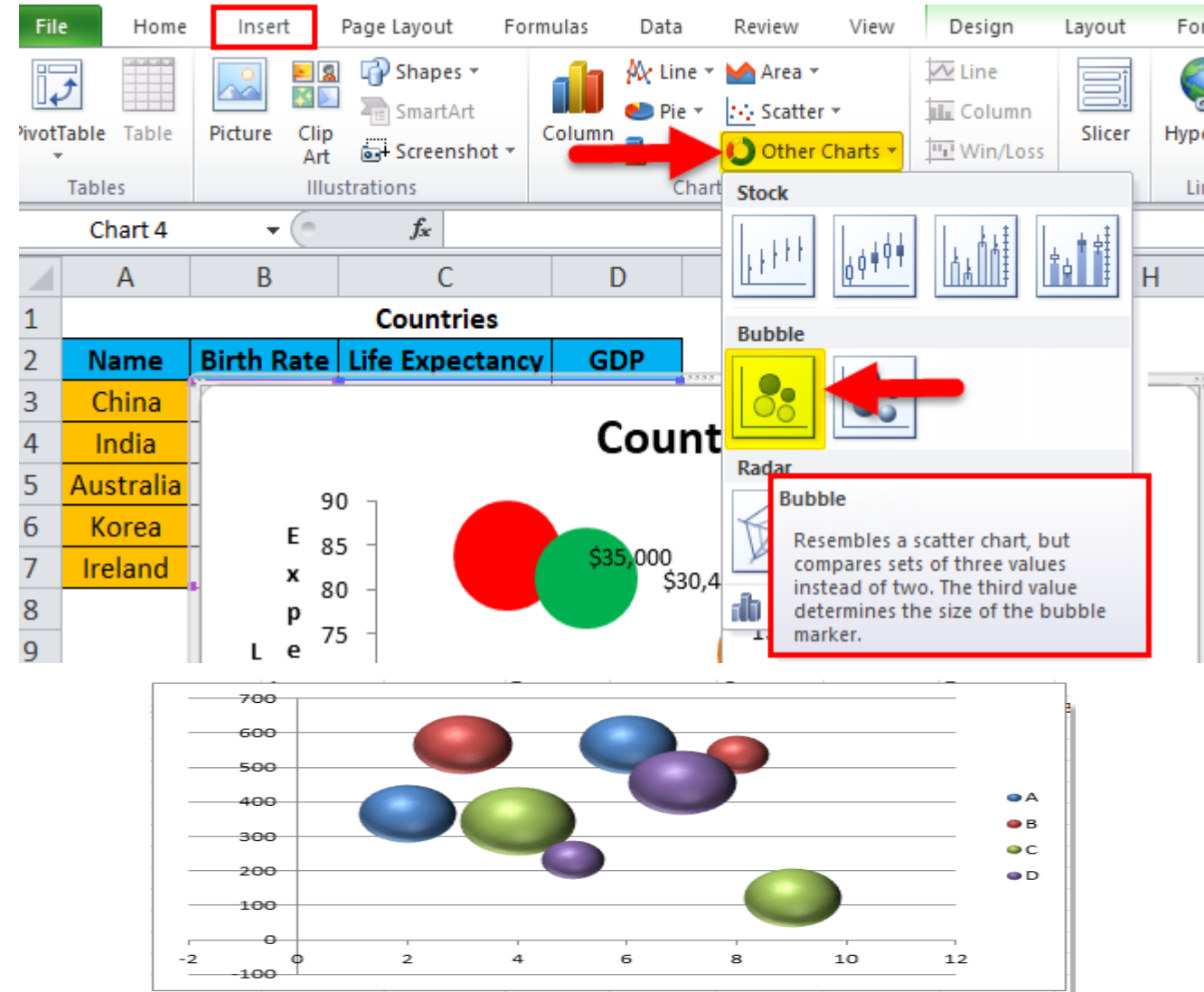


BUBBLE CHART

A Bubble chart is like a Scatter chart with an additional third column to specify the size of the bubbles it shows to represent the data points in the data series.

- A Bubble chart has the following sub-types –
- Bubble
- Bubble with 3-D effect

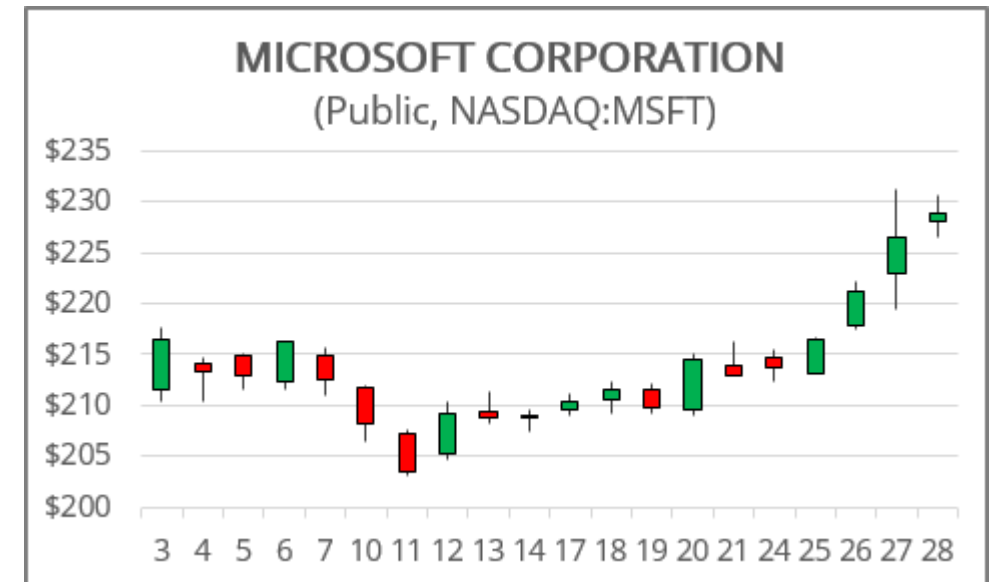
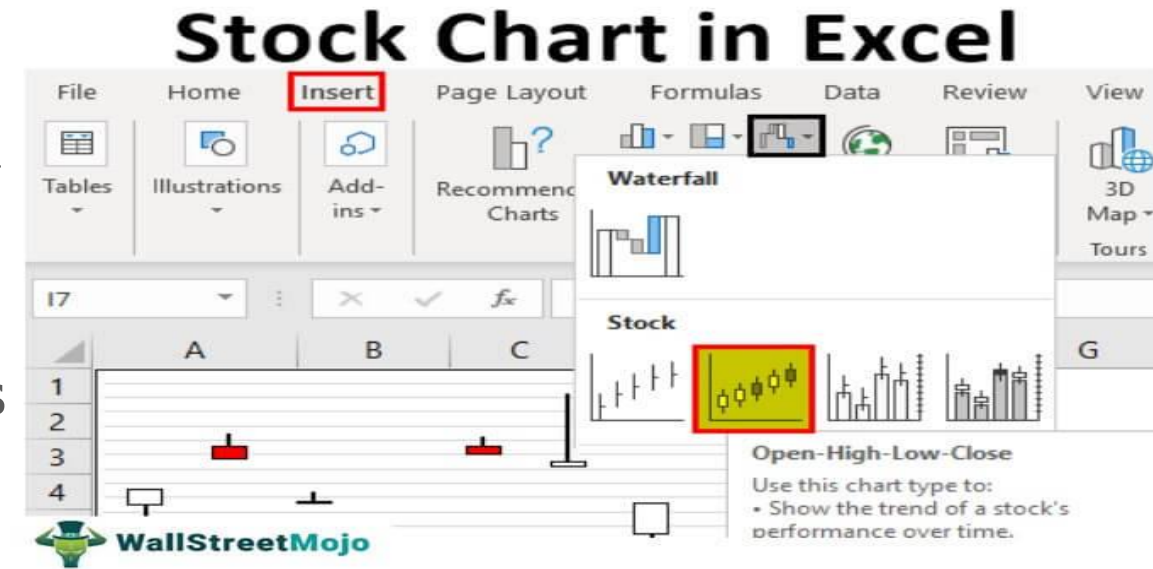
Bubble Chart in Excel



As the name implies, Stock charts can show fluctuations in stock prices. However, a Stock chart can also be used to show fluctuations in other data, such as daily rainfall or annual temperatures.

To create a Stock chart, arrange the data in columns or rows in a specific order on the worksheet. For example, to create a simple high-low-close Stock chart, arrange your data with High, Low, and Close entered as Column headings, in that order.

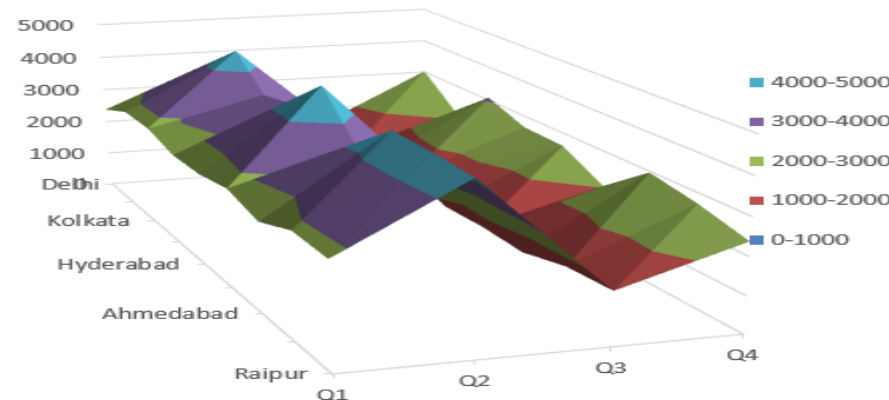
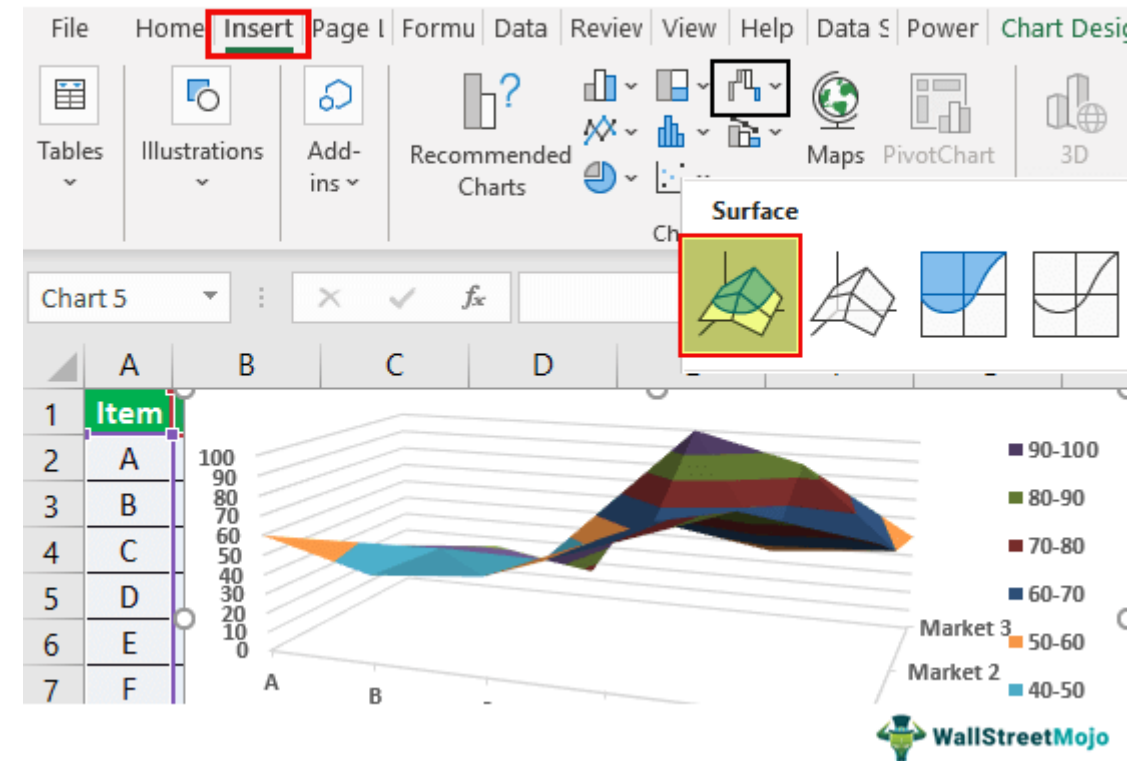
- A Stock chart has the following sub-types –
- High-Low-Close
- Open-High-Low-Close
- Volume-High-Low-Close
- Volume-Open-High-Low-Close



A Surface chart is useful when you want to find the optimum combinations between two sets of data. As in a topographic map, colors and patterns indicate areas that are in the same range of values.

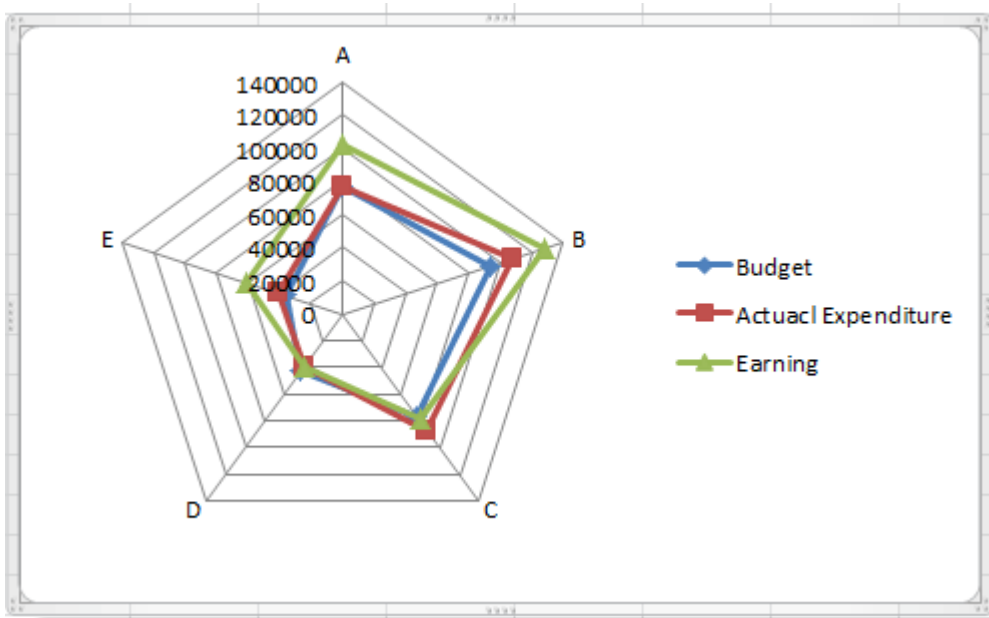
- Ensure that both the categories and the data series are numeric values.
- Arrange the data in columns or rows on the worksheet.
- A Surface chart has the following sub-types –
 - 3-D Surface
 - Wireframe 3-D Surface
 - Contour
 - Wireframe Contour

Surface Chart in Excel

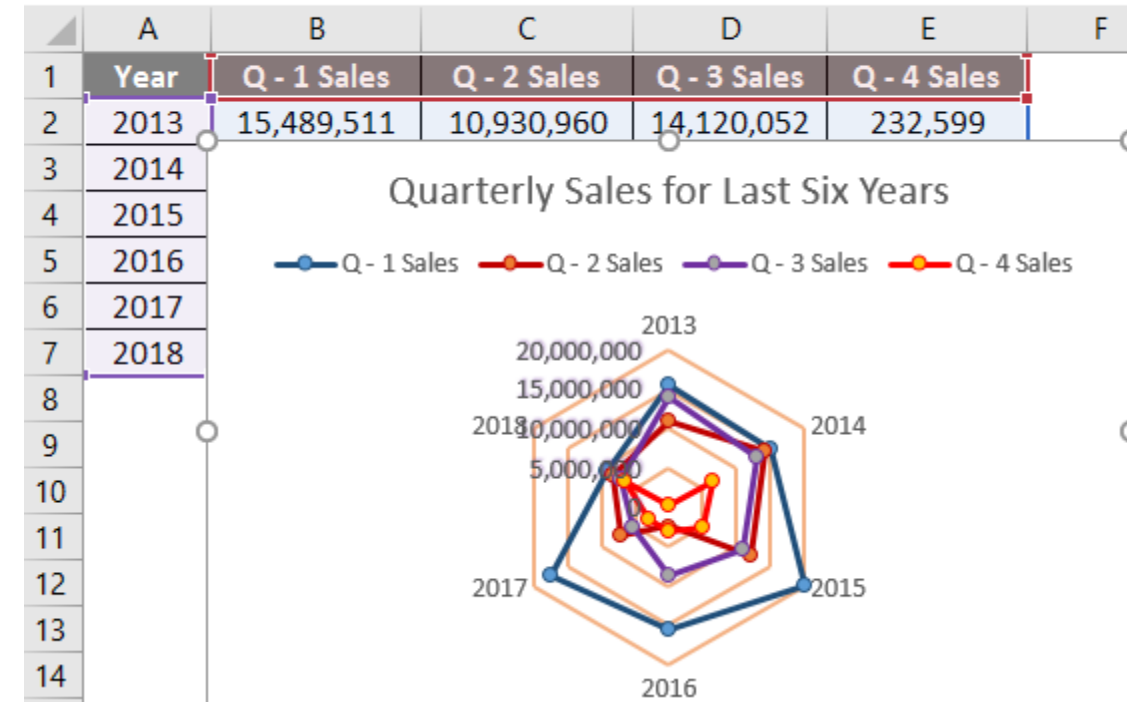


Radar charts compare the aggregate values of several data series. To create a Radar chart, arrange the data in columns or rows on the worksheet.

- A Radar chart has the following sub-types –
- Radar
- Radar with Markers
- Filled Radar



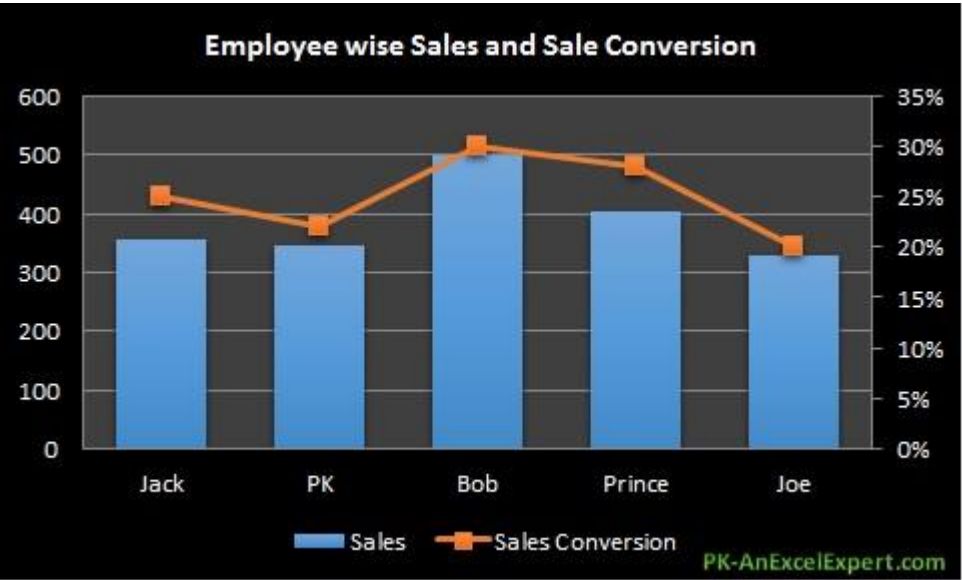
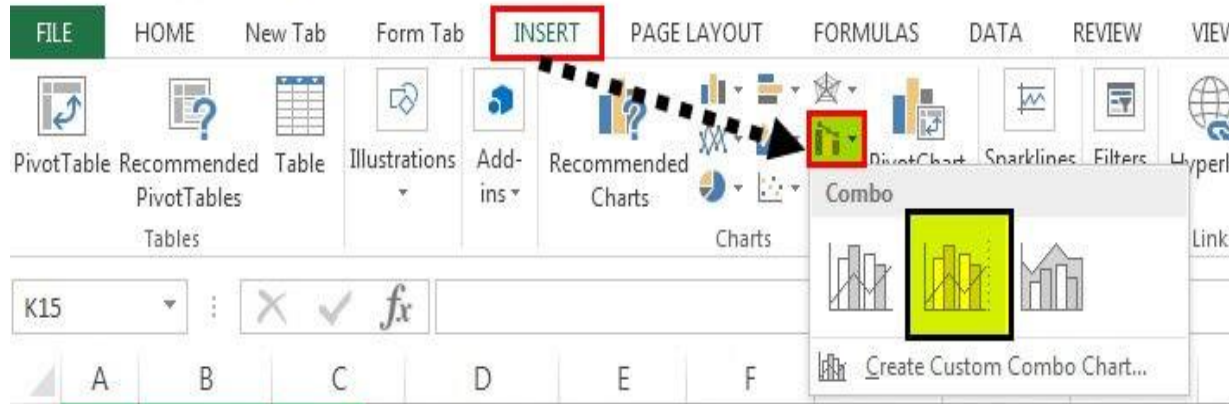
Radar Chart in Excel



Combo charts combine two or more chart types to make the data easy to understand, especially when the data is widely varied. It is shown with a secondary axis and is even easier to read. To create a Combo chart, arrange the data in columns and rows on the worksheet.

- A Combo chart has the following sub-types –
- Clustered Column – Line
- Clustered Column – Line on Secondary Axis
- Stacked Area – Clustered Column
- Custom Combination

Combination Charts in Excel

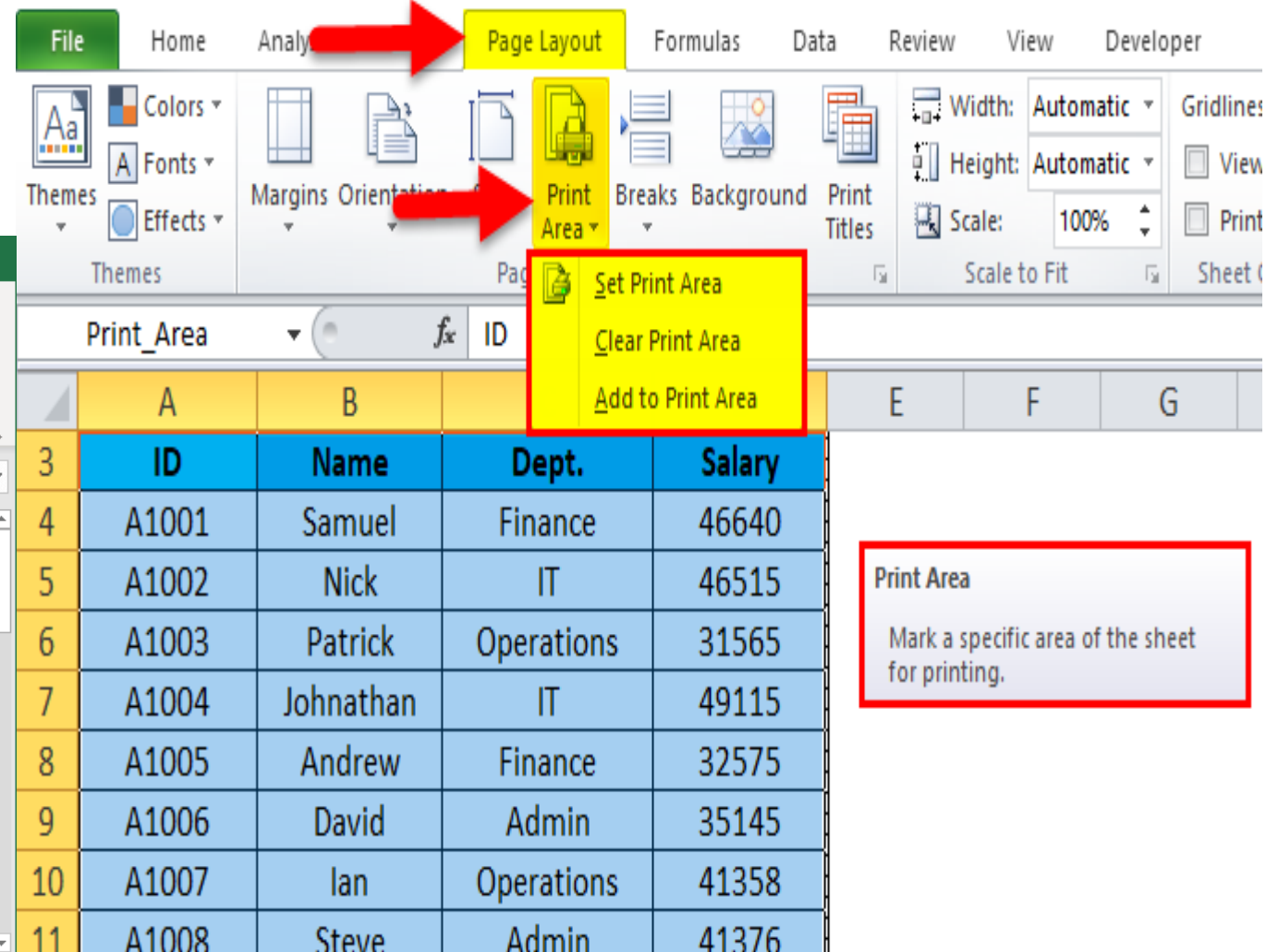


print area, margins, header, footer

On the worksheet, select the cells that you want to define as the print area. Tip: To set multiple print areas, hold down the Ctrl key and click the areas you want to print. Each print area prints on its own page.

On the Page Layout tab, in the Page Setup group, click Print Area, and then click Set Print Area.

Print Area in Excel



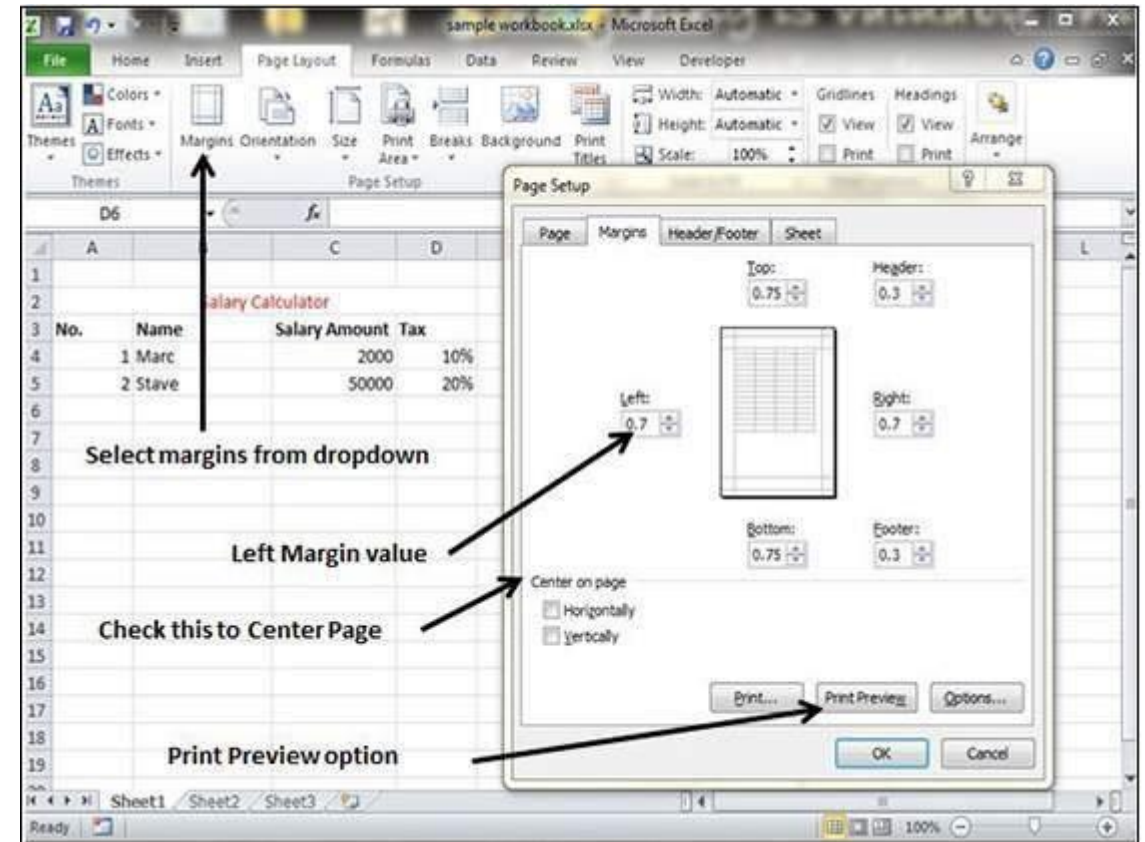
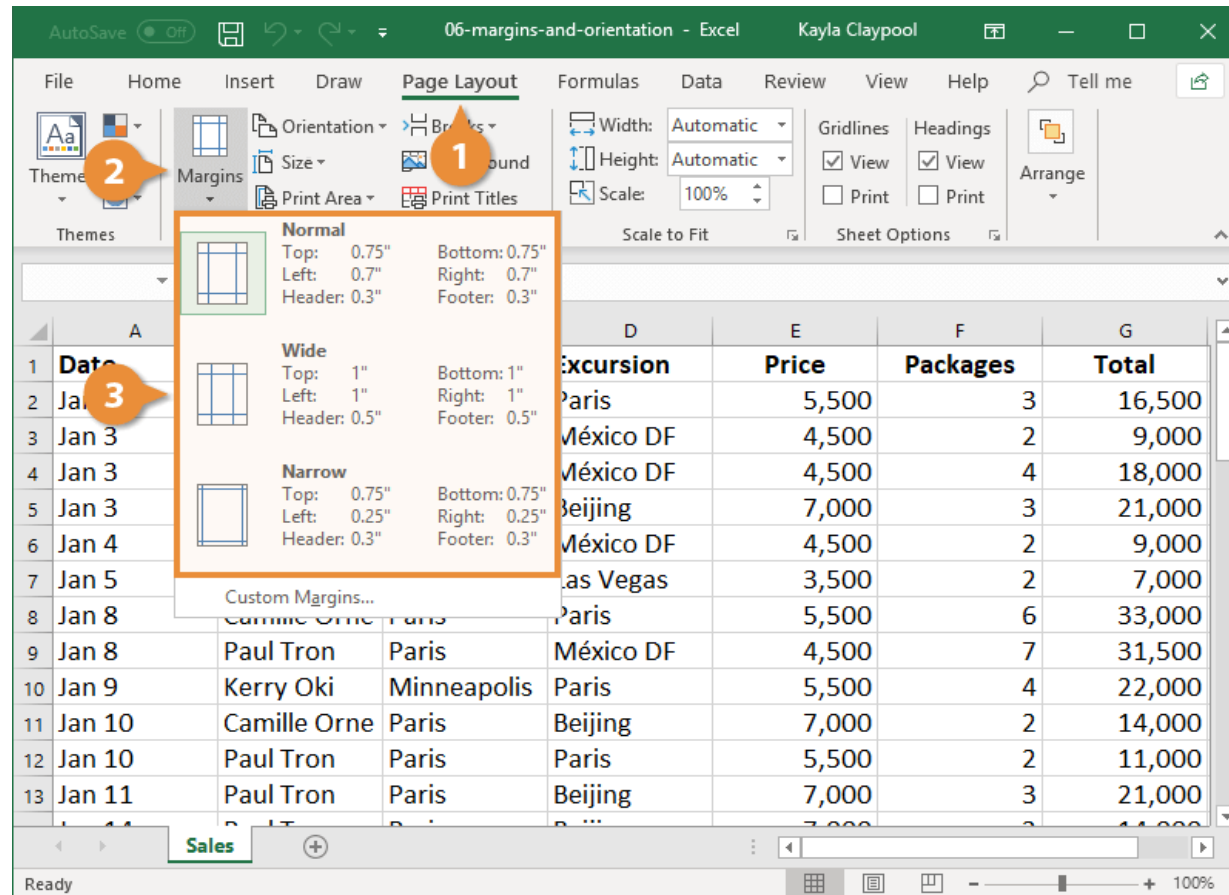
The screenshot shows the Excel interface with the Page Layout tab selected. The Print Area button in the Page Setup group is highlighted with a red arrow. The dropdown menu for Print Area is open, showing three options: Set Print Area, Clear Print Area, and Add to Print Area. A red box highlights this dropdown menu. Another red arrow points to the Set Print Area option. Below the ribbon, a worksheet is visible with a table of employee data.

	A	B	C	D	E	F	G
1	Date	Agent	Office	Excursion	Price	Packages	Total
2	Jan 2	Kerry Oki	Minneapolis	Paris	5,500	3	16,500
3	Jan 3	Kerry Oki	Minneapolis	México DF	4,500	2	9,000
4	Jan 3	Paul Tron	Paris	México DF	4,500	4	18,000
5	Jan 3	Nena Moran	Torreon	Beijing	7,000	3	21,000
6	Jan 4	Kerry Oki	Minneapolis	México DF	4,500	2	9,000
7	Jan 5	Robin Banks	Minneapolis	Las Vegas	3,500	2	7,000
8	Jan 8	Camille Orne	Paris	Paris	5,500	6	33,000
9	Jan 8	Paul Tron	Paris	México DF	4,500	7	31,500
10	Jan 9	Kerry Oki	Minneapolis	Paris	5,500	4	22,000
11	Jan 10	Camille Orne	Paris	Beijing	7,000	2	14,000
12	Jan 10	Paul Tron	Paris	Paris	5,500	2	11,000
13	Jan 11	Paul Tron	Paris	Beijing	7,000	3	21,000

Print Area

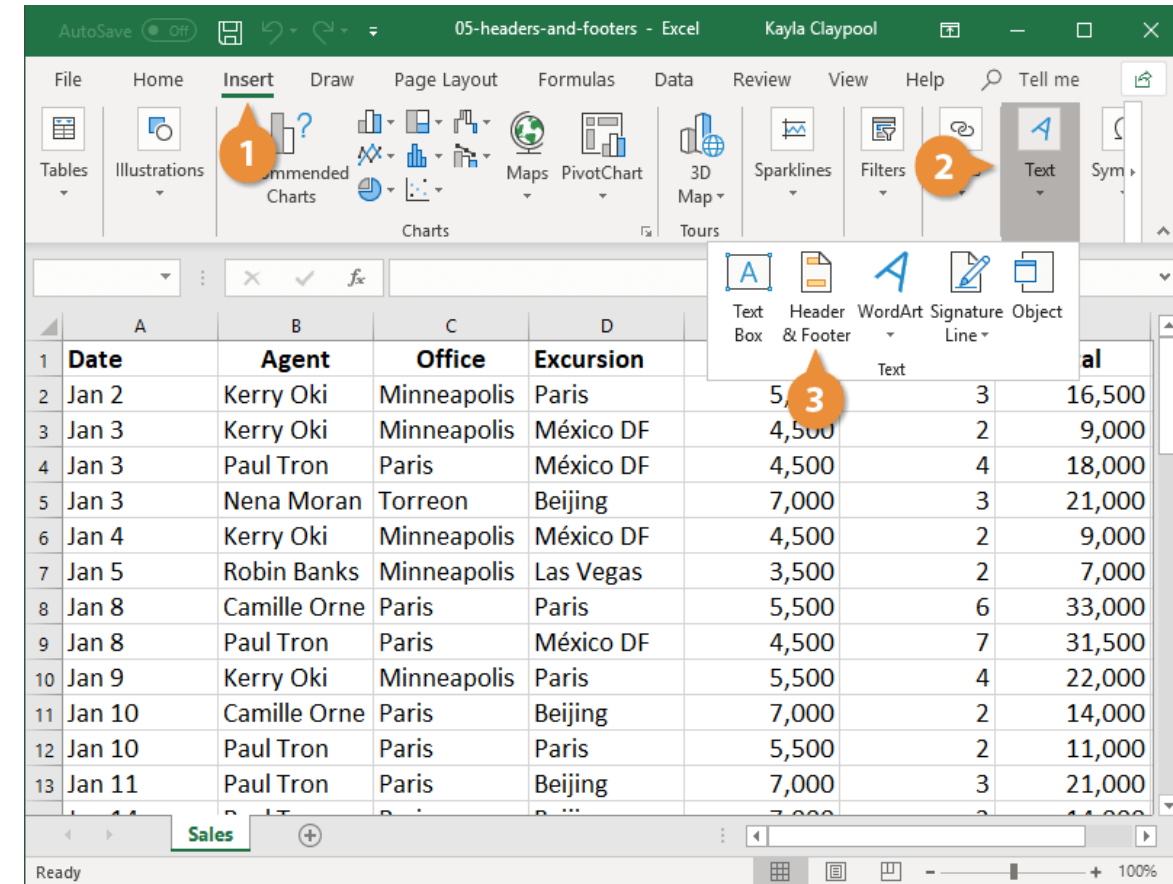
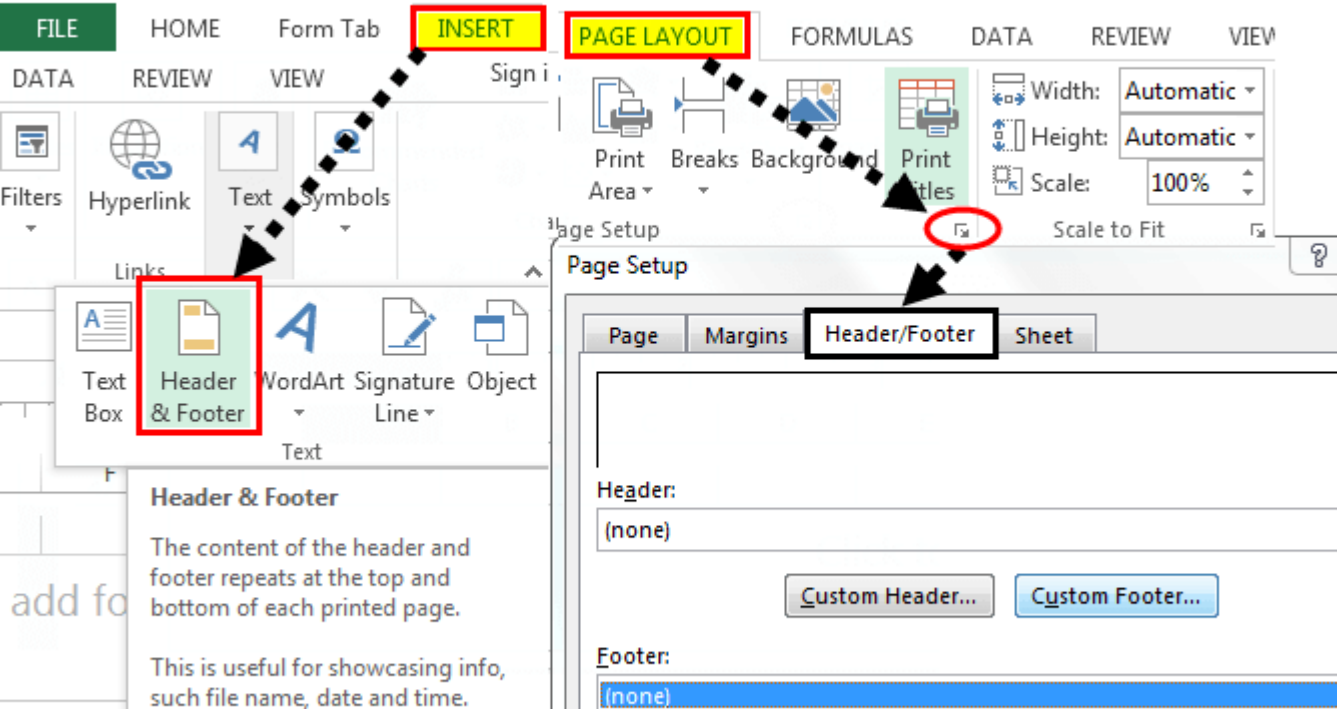
Mark a specific area of the sheet for printing.

- Use the Page Layout tab: On the Page Layout tab, click Margins, and then select Custom Margins.... Use the arrows to increase or decrease the margin sizes, or enter the desired size in the appropriate box. When you're done, click OK



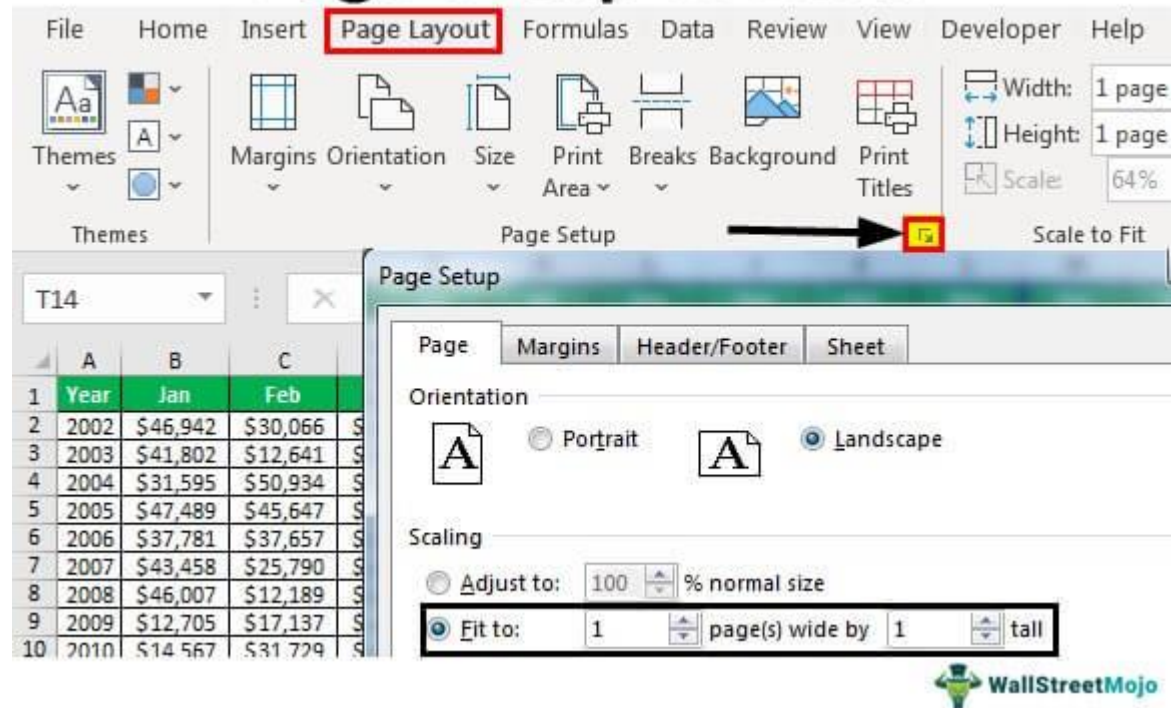
- On the Insert tab, in the Text group, click Header & Footer. Excel displays the worksheet in Page Layout view. To add or edit a header or footer, click the left, center, or right header or footer text box at the top or the bottom of the worksheet page (under Header, or above Footer). Type the new header or footer text.

Header and Footer in Excel



- Press CTRL and then click each worksheet tab in the workbook that you want to affect.
- On the File menu, click Page Setup. Note. In Excel 2007, click the dialog box launcher in the Page Setup group in the Page Layout tab.
- Make the changes that you want in the Page Setup dialog box, and then click OK.

Page Setup in Excel



Page Setup in Excel

This screenshot shows the Excel 2010 interface with the 'Page Layout' tab selected. The 'Page Layout' group is highlighted, and the 'Page Setup' dialog box launcher icon is shown. The 'Page Setup' dialog box is open, displaying the 'Page' tab. The 'Orientation' section shows 'Landscape' selected. The 'Scaling' section shows 'Fit to: 1 page(s) wide by 1 tall' selected. A red box highlights the 'Fit to' settings.

Person wise Sales Data

Sales Rep Name	Quantity	Sales (AED)	Margin
Abul Feizi Syed Mohammed	3,469	-2,099	-2,666,150
Akshay Pramod	8,854	-16,591	-9,083,951
Ambalavanan Agoramurthy	31,577	-128,121	-1,544,251
Anjarappattil Abdul Latheef	6	647,850	110,300

- https://www.tutorialspoint.com/excel_charts/index.htm
- https://www.tutorialspoint.com/excel/excel_simple_charts.htm
- https://www.tutorialspoint.com/excel_charts/excel_charts_quick_guide.htm

THANK YOU